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FEATURE ARTICLE

How easy are the syntheses of allenes?

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Allenes have proven themselves to be valuable building blocks toward complex molecular targets, revealing novel applications in natural product synthesis, pharmaceutical chemistry and materials science. The ongoing interest in allene chemistry results in a variety of new methodologies and pathways for the synthesis of allenes. This feature article highlights some of the recent important developments on the synthesis of allenes and the applications on the synthesis of allenic natural products and allenic-based optoelectronic materials.

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Introduction

In the past few decades, allene chemistry has experienced a veritable explosion.[1] Their synthetic potential in regio- and stereoselective C–C and C–X bond-forming transformations, unique axis-to-center chirality transfer, occurrence of allenic structures in a variety of natural products and pharmacologically active compounds, have inspired ample interest of organic and medicinal chemists. Thus, the synthesis of allenes will surely be a hot research topic for the next decades.[2] All the classical reaction types (addition, elimination, substitution, rearrangement) have been applied to the synthesis of allenes and transition-metal catalyzed allene synthesis have also been developed. However, catalytic stereoselective variations are still challenging and an under-represented area of synthetic organic chemistry.[3]

This feature article highlights some of the recent important developments on the synthesis of allenes with a brief touch on the history, which have not been included in the recent reviews[1n,2e,4] and is organized according to the starting materials and categorized further by reaction types.

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1. From alkenes

1.1 By 1,2-elimination

1,2-Elimination of two groups from a vinyl precursor represents a common class of reactions used to access allenes



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(Scheme 1).[5] For example, allenes can be obtained by dehydrohalogenation of 2-halopropene derivatives using strong bases,[6] *syn*-selenoxide elimination of vinyl selenides,[7] and metal-promoted reduction of 2,3-dihalopropene or β-haloallyl alcohol derivatives.[8] Recently, 2,3-dichloro-1-propene eliminating two molecules of KCl in the presence of K_2Te_2 to afford propadiene has also been reported.[9]

This elimination protocol has been recently applied to the synthesis of 1,1-difluoroallenes (Scheme 2).[10] Upon treatment with *n*-butyl lithium, 2-bromo-3,3-difluoroallylic acetates **1** underwent lithium-bromide exchange, followed by 1,2-elimination of lithium acetate to afford mono- and disubstituted 1,1-difluoroallenes **2** in high yields. The corresponding starting material could be prepared conveniently from the reaction of aldehydes or ketones with 1-bromo-2,2-difluorovinyl lithium. The use of hexane was essential since it is a nonpolar solvent resulting in a higher degree of aggregation of lithium reagents. Its low reactivity helped to suppress undesired side reactions. Similarly, Yamazaki *et al.* used zinc-mediated 1,2-elimination of 2-iodo-3-trifluoromethylallylic acetates to prepare the potentially important CF_3-substituted allenes.[11]

Based on Knochel's Mg–Br exchange protocol, Wu *et al.* have reported an enantioselective synthesis of 1,3-disubtituted allenes **5** from optically active ethyl 6-acetoxy-5-iodotetradeca-2,4-dienoate **4** in good yield and a high level of stereoselectivity (Scheme 3).[12]

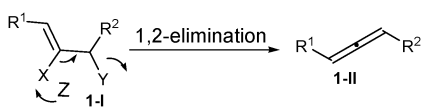
The oxidative *syn*-elimination reaction of (*Z*)-3-phenylseleno-3-alkenyl methyl ethers **6** with *m*CPBA furnished functionalized allene derivatives **7** in good yields (Scheme 4).[13] The configuration of the C=C bond was important for the formation of allenes since the corresponding selenoxide elimination of (*E*)-3-phenylselenohomoallylic ethers exclusively gave alkyne derivatives.

The reaction of 1-chlorovinyl sulfoxides **9** with *i*PrMgCl gave a magnesium species **11** *via* sulfoxide-magnesium exchange (Scheme 5).[14] Treatment of this magnesium carbenoid with 2-lithio-5-methoxyfuran resulted in the formation of functionalized allenes **10** in moderate yields.

Furthermore, when magnesium species **15** were trapped by lithium enolates of α-chloroalkanoates, fully substituted 2,3-allenonates **14** could be obtained in 17–51% yield (Scheme 6).[15]

Treatment of β-ketoesters **17** with 2 equiv. of LiHMDS and Tf_2O formed vinyl enolate intermediates **19**. Subsequent addition of another equivalent of LiHMDS, 3 equiv. of HMPA, and $ZnCl_2$ led to terminal 2,3-allenoates **18** in good yields (Scheme 7).[16]

Surprisingly, Bertrand *et al.* synthesized the highly strained cationic 4-membered allenes **21** bearing two piperidino groups by alkylation of 3-ethoxyspiro[3,5]non-2-en-1-one **20** with $Et_3O^+BF_4^-$, followed by addition of excess piperidine (Scheme 8).[17] Upon lithiation at −20 °C, cyclic allenes **22** were cleanly produced. This compound is highly reactive and stable only for several hours at −20 °C. This strategy has also been applied to the synthesis of 5-membered cyclic allenes **24** and **25** (Scheme 9).[18]



Scheme 1

1.2 By intramolecular eliminative H-migration

Upon treatment of the tosylhydrazone derivatives **28** generated *in situ* from the reaction of α,β-unsaturated enones **26** with tosylhydrazine, allenes **27** were produced with loss of N_2 and intramolecular H-transfer of the diazene intermediate **29** (Scheme 10).[19]

1.3 By Wittig and related reactions

Wittig reaction is an important method for the formation of C=C bonds.[20] The reaction of phosphonium ylides and ketenes has been employed to access a variety of functionalized allenes, including allenyl esters,[21] ketones,[22] lactones and lactams *etc*.[23] To avoid the manipulation of unstable ketenes, their equivalents, such as acid halides[24] or aryl esters,[25] are commonly used.

Palladium-catalyzed allylic alkylation of phosphorus ylides **30** in the presence of a Pd/Trost ligand system afforded a new class of allylic phosphorus ylides **34**, which reacted with various ketenes furnishing tetrasubstituted allenes **32** in moderate to good overall yields (Scheme 11).[26]

Petasis olefination, involving conversion of carbonyl compounds to olefins, is an important process for construction of C–C double bonds in organic synthesis.[27] The use of alkenylidene titanocene provides a direct and versatile way to prepare a variety of allenes from aldehydes or ketones.[28] Recently, Zhou *et al.* reported a tandem reaction starting from allylgallation of magnesium acetylides, followed by the reaction with aldehydes to afford allyl allenes **37** in moderate yields (Scheme 12).[29]

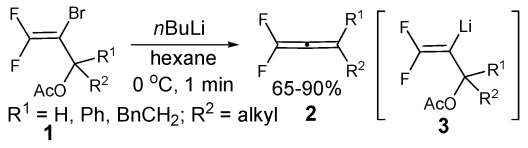
When α-silyl α-diazo ketones **39** and silyl ketenes **40** were combined at 0–20 °C for 3–12 h, 1,3-diaryl-3-silyl-1-silyloxy-allenes **41** were formed in fairly good yields (Scheme 13).[30] The reaction might proceed *via* cyclopropanonation of silyl ketenes **40** with 1-diazo-2-silyloxyethenes **42**, followed by decarbonylation to yield 1,3-diaryl-1-silyloxy-3-silylallenes **41**.

1.4 By C–C bond cleavage

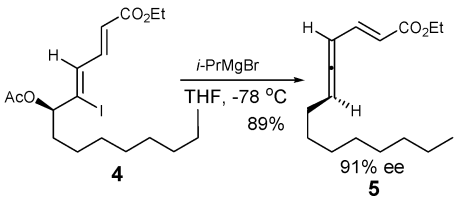
C–C fragmentation has been less explored in the preparation of allenes. In 2005, Nagao *et al.* developed a strategy for the synthesis of chained keto allenoates by retro-Dieckmann-type[31] or retro-aldol-type[32] reactions of α-alkynyl-α-ethoxy-carbonyl cyclopentanone and cyclohexanone derivatives under basic conditions. Recently, Williams and co-workers have shown that β-siloxy substituted vinyl triflates **44** smoothly underwent C–C fragmentation to afford trisubstituted allenes **45** stereoselectively in the presence of TBAF (Scheme 14).[33]

Nucleophile-induced Grob-type fragmentation[34] reactions of vinylogous acyl triflates **47** leading to β-functionalized allenes **48** were developed by Cramer *et al.*[35] The reaction led to trisubstituted allenes with carbonyl functionality at the oxidation level of a carboxylic acid (Scheme 15).

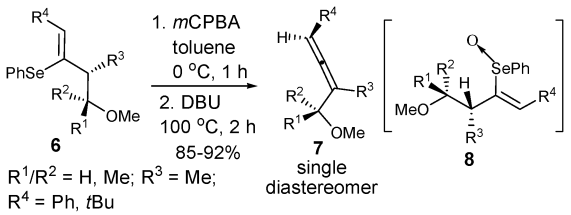
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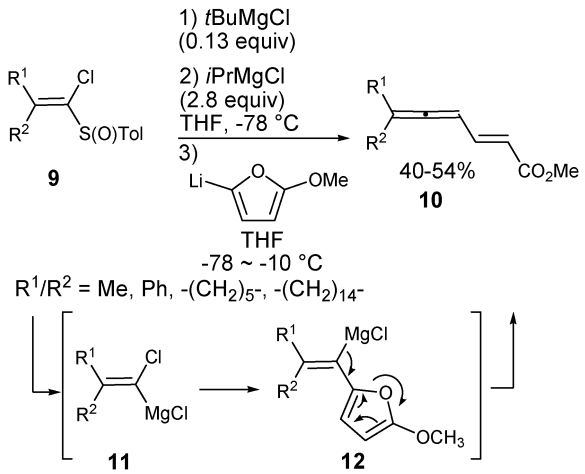
Scheme 2



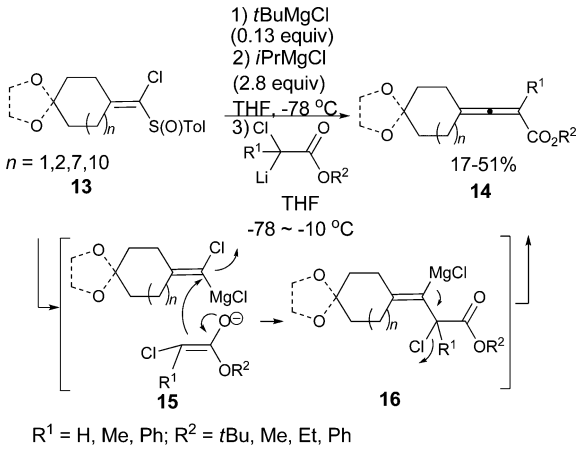
Scheme 3



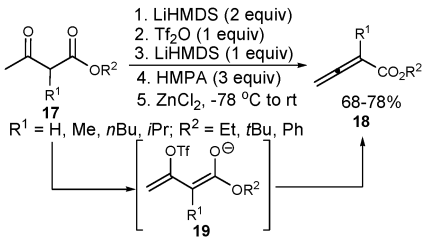
Scheme 4



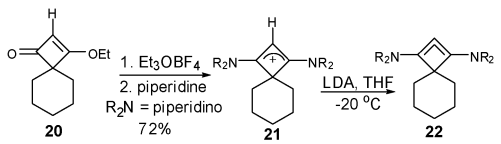
Scheme 5



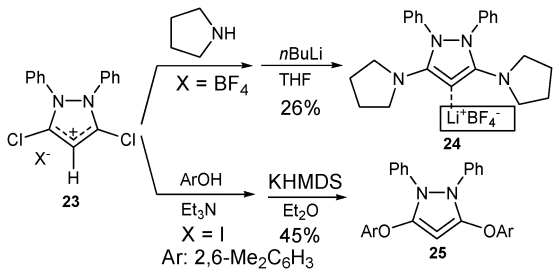
Scheme 6



Scheme 7



Scheme 8



Scheme 9

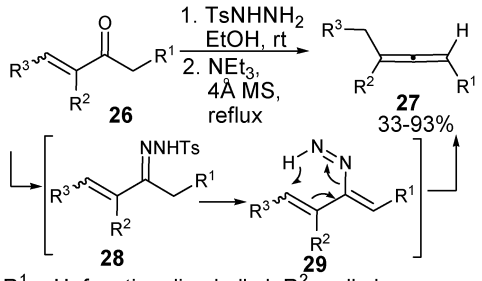
1.5 By allylic substitution of 2-halo-1,3-butadienes

The transition metal-catalyzed reactions of 2-halo-1,3-dienes with various nucleophiles, such as sodium malonates, sodium phenoxides, or potassium amides, provide a highly selective and practical tool for the synthesis of α-functionalized allenes (Scheme 16).[3b,36] The most commonly used transition metals are organocopper and palladium(0) complexes. For the former, organocuprates usually act as nucleophiles to react with 2-halo-1,3-butadienes **1-III** *via* S_N2' or S_N2 processes furnishing allenes **1-VI** or 1,3-butadiene derivatives **1-V**, respectively. As for the latter, (alkylidene-π-allyl)palladium is the key intermediate. Attacking the 1- or 3-position of this intermediate affords the corresponding allenes **1-VI** or 1,3-dienes **1-V**. Many factors such as solvents, temperature affect the regioselectivities. Highly enantioselective preparation of axial chiral allenic compounds could also be achieved by the use of chiral phosphines.[37]

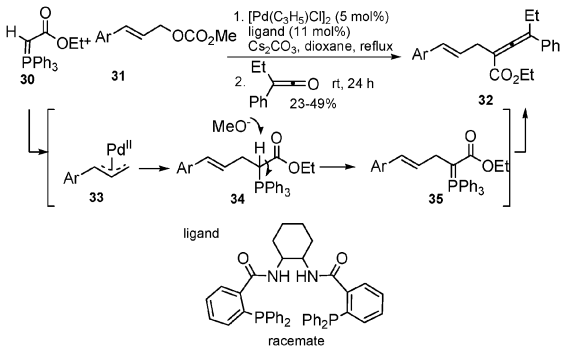
Recently, this methodology was extended to the preparation of fluorobis(phenylsulfonyl)methylated allenes **52** from 2-bromo-1,3-dienes (or a related triflate) **50** and fluorobis(phenylsulfonyl)-methane **51** (Scheme 17).[38] The fluorobis(phenylsulfonyl)methyl

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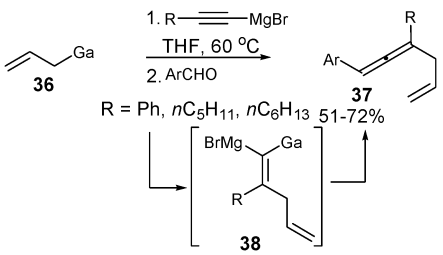
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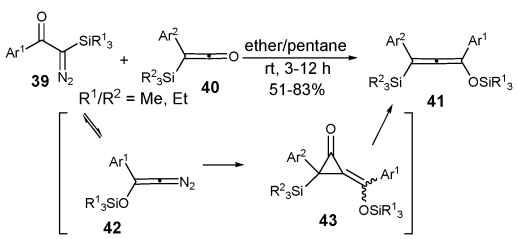
Scheme 10



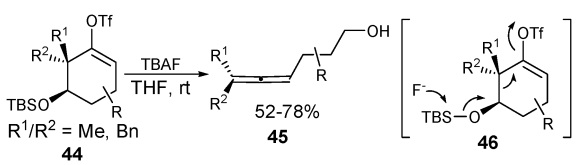
Scheme 11



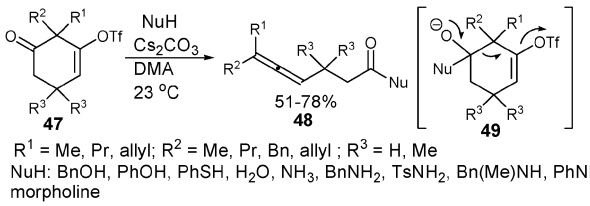
Scheme 12



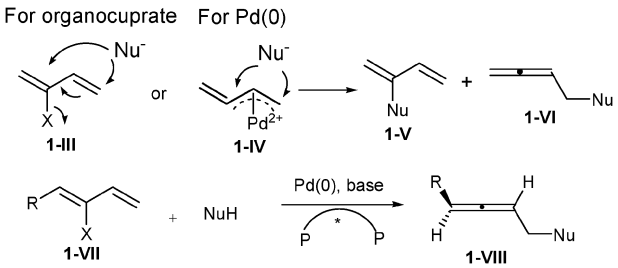
Scheme 13



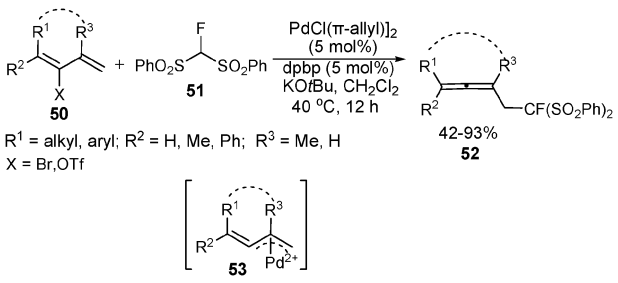
Scheme 14



Scheme 15



Scheme 16



Scheme 17

group underwent reductive desulfonylation with activated Mg turnings in MeOH–THF (1/3) affording the β-fluoroallenes in high yields.

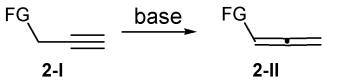
2. From alkynes

Alkynes, isomers of allenes, are probably the most widely used starting materials for the synthesis of allenes.[39]

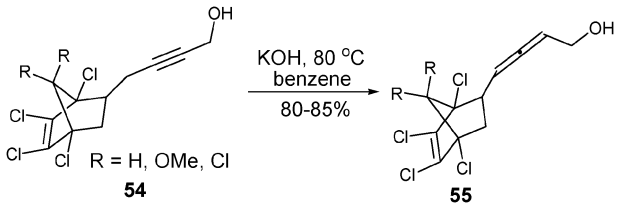
2.1 By isomerization of alkynes

Isomerization of the FG–CH_2–C≡C moiety in the presence of a base is one of the earliest methods for access to allenes (Scheme 18).[40] Simple alkynes are usually treated with strong bases such as alkali metal amide ($NaNH_2$) at a high temperature.[41] For alkynes with activated α-hydrogen or coordination to a transition metal, weaker bases such as NaOH, KO*t*Bu and TBAF are enough to facilitate the isomerization.[42] The activation groups include alkene,[43] arene,[44] alkyne,[45] or carbonyl group,[46] or groups containing oxygen,[47] nitrogen,[48] sulfur,[49] or phosphorus atom.[50]

Polychloro bridged bicyclic propargyl alcohols **54** isomerized to the corresponding allenols **55** in the presence of powdered KOH in benzene at 80 °C with 80–85% yields (Scheme 19).[51]



Scheme 18



Scheme 19

Propargylic tosylates **56** with the CF_3 group at the terminal *sp* carbon atom isomerized to allenyl tosylates **57** smoothly at room temperature in the presence of sodium hydroxides in a water–organic biphasic solvent system (Scheme 20).[52]

Quite excitingly, an enantioselective isomerization of 3-alkynonates **58** to optically active 2,3-dienonates **60** with chiral bicyclic guanidine **59** as base was also developed (Scheme 21).[53]

Oxidation/isomerization of homopropargyl alcohols is a convenient way for synthesis of α-allenyl ketones.[54] Recently, Fan *et al.* reported a green catalytic system using $RuCl_3$ in ionic liquid with *tert*-butyl hydroperoxide (TBHP) as the oxidant (Scheme 22).[55]

The coupling reaction of α-methyl benzylbromide **63** with the trialkynylindium derivative of ethyl propiolate **64** in the presence of chiral nickel complex afforded axially chiral allene **65** in 30% yield with good enantioselectivity (77% ee) (Scheme 23).[56] Other alkynylindium derivatives with aryl, vinyl and TMS substituents gave benzyl-substituted alkynes.

2.2 By S_N2'-substitution

Substitution reaction of propargylic compounds with a leaving group is the most commonly employed and convenient method for the synthesis of allenes (Scheme 24).[57] In many cases, non-racemic propargylic substrates proceed in a stereoselective manner. A variety of labile groups, including propargylic acetates, mesylates, carbonates, and halides, *etc.* have been used as leaving groups.[58]

2.2.1 Substitution by nucleophile reagents. Haloallenes are useful substrates for substitution reactions[59] and transition metal-catalyzed cross-coupling reactions.[60] They have also been found in many natural products.[61] A variety of reaction conditions for the preparation of haloallenes from propargyl alcohols have been developed.[62]

Treatment of aryl-substituted propargyl alcohols **66** with *N*-halosuccinimide and triphenylphosphine afforded chloro- or bromoallenes **68** in moderate to good yields and high regioselectivities (Scheme 25).[63]

Appel-type reaction conditions containing Ph_3P, CBr_4, $(n\text{-}Bu)_3P$, and $i\text{-}Pr_2NEt$ were also applied to the bromination of 1-arylpropargylic alcohol **69** affording allenyl bromides **70**.[64] Bromination of CF_3-substituted propargylic alcohols **71** by the combined reagents CBr_4–Ph_3P gave bromoallenes **72** (Scheme 26).[65]

Haloallenes can also be prepared from propargyl boronates (Scheme 27).[66] Treatment of propargyl diethanolamine boronates **73** with *N*-halosuccinimides **74** gave the corresponding allenyl bromides **75** in high yields.

The reaction of secondary or tertiary propargylic alcohols **76** with $NbBr_5$ afforded allenylic bromides **78** in moderate to good yields (Scheme 28).[67] The halides were readily isolated in pure form through simple extraction and could be used directly in subsequent transformation.

Synthesis of optically active bromoallene **80** by treatment of enantioenriched propargyl tosylate or mesylate **79** with LiBr and CuBr in THF was developed by Montury and Goré in 1980.[68] During Wu's recent investigations,[69] they noticed that the enantiopurity of the bromoallenes could change dramatically from run to run (Scheme 29). After systematic screening on the reaction conditions, they found that Cu(II) species played an unexpected role in improving the enantioselectivity in this particular transformation. Commercially available 'anhydrous' diethyl ether (with H_2O content $<0.3\%$) was a much better solvent than THF, which was the only solvent that has been used for this reaction up to now. The newly established conditions (addition of 0.1 equiv. of $Cu(OTf)_2$ with Et_2O to replace THF) increased the regioselectivity from 1.5 : 1 [S_N2'/S_N2] to 3.8 : 1 and the er from 13 : 1 (85% ee) to 46 : 1 (96% ee).

DBU reacts with CO_2 to form DBU–CO_2 zwitterionic carbamic complex **83**, which has been observed for various carbon dioxide fixation reactions.[70] Yoshida and Ihara *et al.* applied this species to the synthesis of oxazolidinones starting from 4-(benzylamino)-2-butynyl carbonates and benzoates **82** (Scheme 30).[71] 5-Vinylideneoxazolidin-2-ones **85** were obtained in good yields under palladium-free conditions.

The reaction of 2-aryindoles **86** with tertiary propargylic alcohols **87** in the presence of Brønsted acid gave 3-allenylindole derivatives **88** in moderate to good yields (Scheme 31).[72] The competitive isomerization of the resulted allenes to dienes process could be slowed down when the substituents at the propargylic position is secondary alkyl groups. 1,3-Dicarbonyl derivatives reacted with tertiary propargylic alcohols also giving similar results.[73]

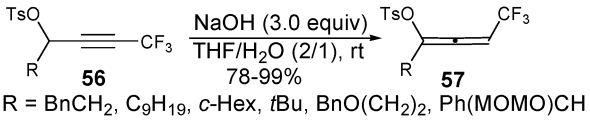
Syntheses of allenes from propargylic alcohols can also be realized by using appropriate activating reagents. For example, phosphane-like PPh_3 was employed as a nucleophile and deoxygenation reagent efficiently to promote the transformation of 2,4-alkadiynols **89** into allenynes **90** (Scheme 32).[74] The hydrogen was proposed to transfer intramolecularly from the hydroxyl group. The presence of a conjugated alkynyl group is necessary for the conversion of a propargyl alcohol into an allene, as propargyl alcohols with only one alkynyl group resulted in no reaction with the recovering of the starting materials.

One efficient access to allenes is base-mediated desilylation of propargyl silanes.[75] With 1.5 equiv of TBAF in THF at room temperature, the TMS-substituted propargyl alcohols **91** were transformed into the 2,3-allenols **92** in good to excellent yields (Scheme 33).[76]

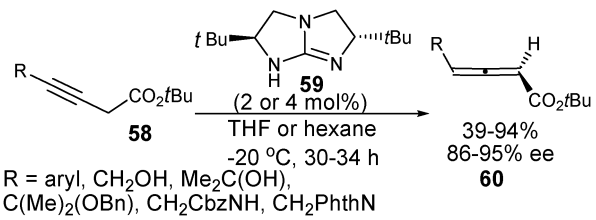
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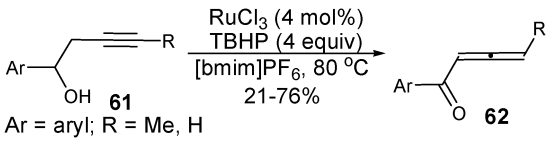
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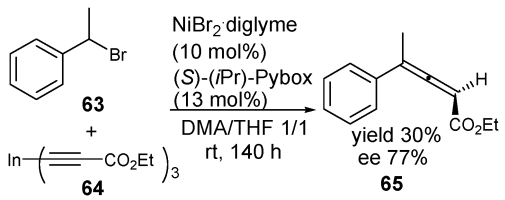
Scheme 20



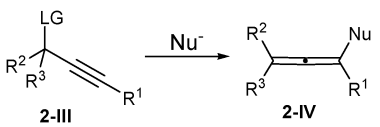
Scheme 21



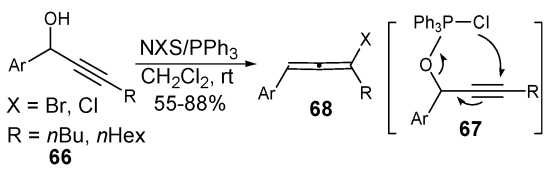
Scheme 22



Scheme 23



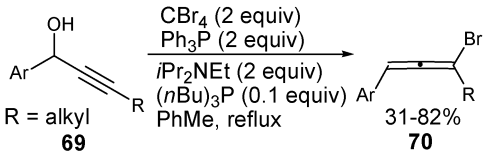
Scheme 24

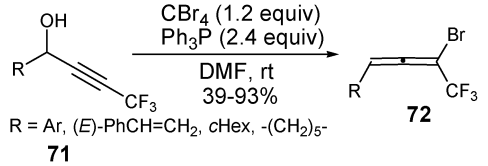


Scheme 25

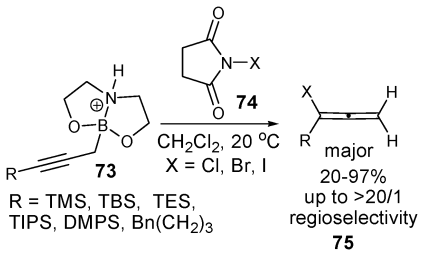
2.2.2 Substitution with organometallic reagents. Organocopper-mediated S_N2' substitution reactions are one of the most versatile and popular synthetic routes to allenes.[58a,b,77] The easily available diorganocuprates (Gilman's cuprates) are commonly used reactive copper reagents. Catalytic variants involving the use of Grignard reagents with a small amount of copper(I) salt have also been developed, which avoids the usually observed formation of acetylenic derivatives resulting from direct S_N2 nucleophilic substitution.[78]

2.2.2.1 Stoichiometric organocopper reagents as nucleophiles. Trifluorovinyl, perfluorophenyl or (*Z*)-pentafluoropropenyl copper reagents **93** readily reacted with tertiary propargylic chlorides **94** to give perfluoro-substituted allenes **95** stereoselectively in good yields (Scheme 34).[79]

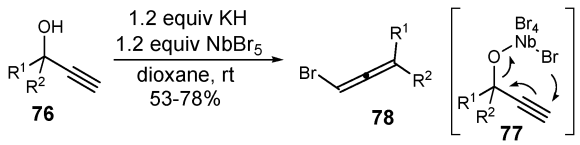




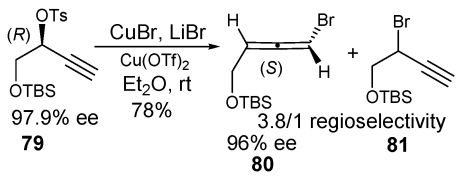
Scheme 26



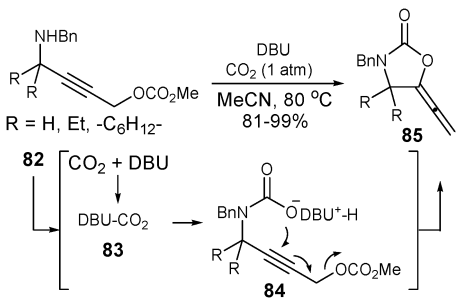
Scheme 27



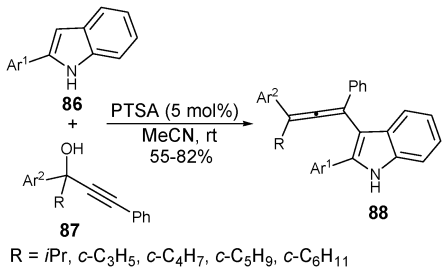
Scheme 28



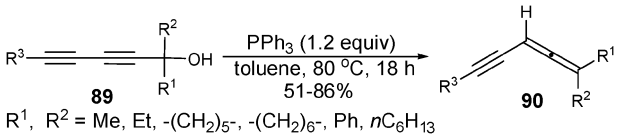
Scheme 29



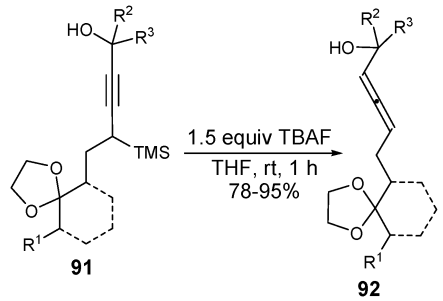
Scheme 30



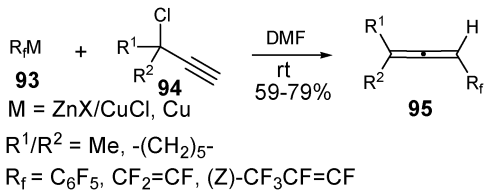
Scheme 31



Scheme 32



Scheme 33



Scheme 34

Enantiomerically enriched allenyl silanes **97** have been produced stereoselectively by treatment of enantiopure oxazolidine propargyl acetate **96** with (trimethylsilyl)cyanocuprate **98** *via* an anti-S_N2' substitution process (Scheme 35).[80]

Treatment of propargyl bromides or mesylates **100** with a chiral sulfoxide **99** containing benzylic copper complexes afforded the chiral allenes **101** by an anti S_N2' process (Scheme 36).[81] The sulfinyl group efficiently transferred its chiral information to the products. When optically active propargyl mesylates were subjected to the reaction, up to 95/5 diastereoselectivity was observed.

2.2.2.2 Copper(I)-catalyzed nucleophilic substitution. Cu-catalyzed nucleophilic substitution reactions exhibited higher yields and selectivities than analogous stoichiometric amounts of Cu(I) salt-mediated reactions. Commonly used organometallic reagents included organolithium,[82] Grignard reagents,[83] organozinc[84] and organozirconium,[85] *etc.*

Groth *et al.* developed an approach towards enantiomerically enriched α-allenylglycines **105** utilizing homocuprate of the bislactim ether of *cyclo*-(–L-Val–Gly–) **102** and primary propargyl halides **103** (Scheme 37).[86] This copper-catalyzed S_N2' substitution produces the allenyl-substituted bislactim ethers **104** in a highly diastereoselective manner, albeit, the yields are not high due to the formation of the S_N2 substitution products. Subsequent hydrolysis of the allenyl-substituted dihydropyrazines **104** affords the α-allenylglycines **105**.

Copper(I)-catalyzed anti-S_N2'-type reduction of propargylic carbonates with hydrosilanes to various di- and trisubstituted allenes have been reported.[87,88] NHC ligands (Scheme 38)[87b] or phosphine ligands (Scheme 39)[88] were used to stabilize the corresponding CuH complexes. These reactions have a good tolerance to various functional groups and work efficiently for the synthesis of optically active allenes.

When the hydride source was replaced by *bis*-pinacolato diboron **113**, various allenylboronates **114** could be afforded efficiently. The Cu(I)/Xantphos system also worked well for the preparation of axially chiral allenylboronate **116** from an optically active propargylic carbonate **115** (Scheme 40).[89]

Propargylic derivatives bearing an alkoxy leaving group have been used for access to polyalkylpropa-1,2-dienes,[78a] 1,2,4,6-tetraenes[90] and α-allenylamines.[91] For synthesis of chiral 2,3-allenols, optically active 4-hydroxy-4-aryl-2-butynyl methyl ethers **117** were subjected to primary alkyl Grignard reagents under the catalysis of CuBr (Scheme 41).[92] Enantioenriched 2-substituted secondary 2,3-allenols **118** were obtained with up to >99% ee. However, the yields were somewhat low and the 4-substituent groups were limited to aryl groups.

Then, the related propargyl chlorides **119** were employed to overcome those shortcomings for the above mentioned copper(I)-catalyzed nucleophilic substitution (Scheme 42).[93]

2.2.2.3 Organocopper-mediated ring-opening reactions. The organocopper-mediated ring-opening reaction of alkynylic epoxides,[82,94] lactones[95] or cyclic carbonate[96] affords the corresponding allenols in high regio- and *anti*-diastereoselectivity. Recently, this strategy was applied to the synthesis of bisallenols **122** (Scheme 43).[97] Addition of Grignard reagents with a catalytic amount of CuBr to bis(acetylenic epoxides) **121** led to conjugated bisallenols **122**. The reaction proceeded in two steps and monosubstitution products could be isolated at very low temperatures.

Treatment of acetylenic dioxolanones with organomagnesium cuprate reagents afforded α-allenols in good to excellent yields (Scheme 44).[98]

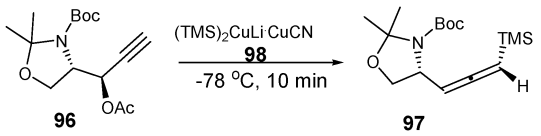
2.2.3 Substitution by other organometallic reagents.[99] Allyl indium reagents, generated *in situ* from indium and allyl bromides **127**, reacted with tertiary propargyl alcohols **126** affording tri- and tetra-substituted allenes **128** in good yields (Scheme 45).[100]

When propargyl bromide **129** and chlorosilane were treated with 1 equiv. of indium powder in DMF at room temperature, allenylsilane **130** were produced highly selectively (Scheme 46).[101]

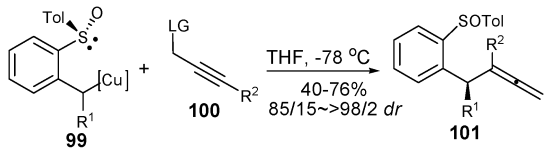
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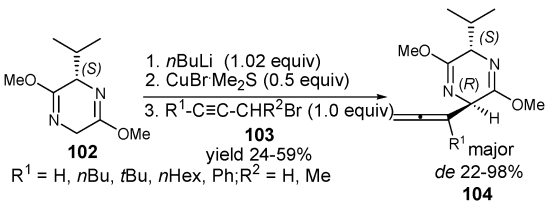


Scheme 35

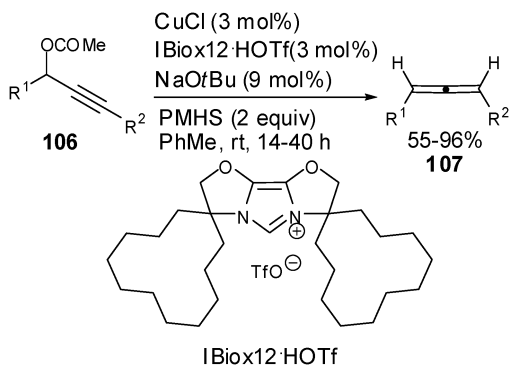


R^1 = Me, Bn, Allyl; R^2 = H, Me, Ph
LG: Br, OMs;

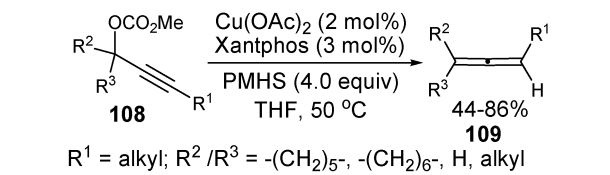
Scheme 36



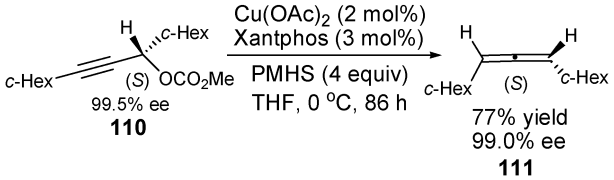
Scheme 37



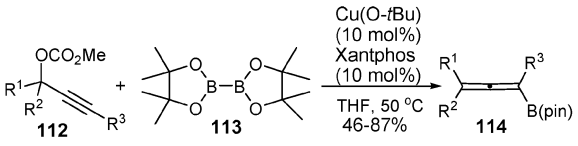
Scheme 38



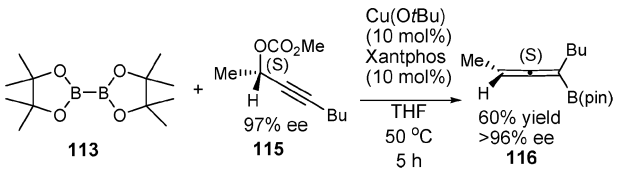
R^1 = alkyl; R^2 /R^3 = -$(CH_2)_5$-, -$(CH_2)_6$-, H, alkyl



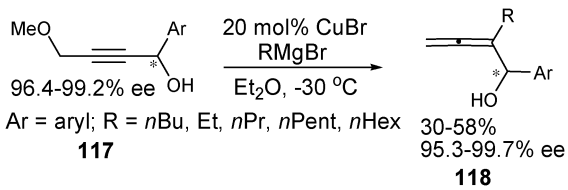
Scheme 39



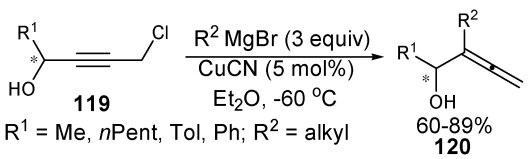
R^1, R^2 = H, Me, $BnCH_2$, $PhCO_2(CH_2)_3$, -$(CH_2)_5$-, -$(CH_2)_6$-; R^3 = alkyl



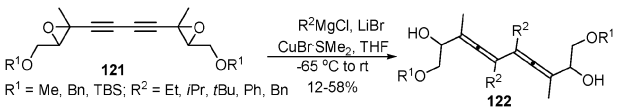
Scheme 40



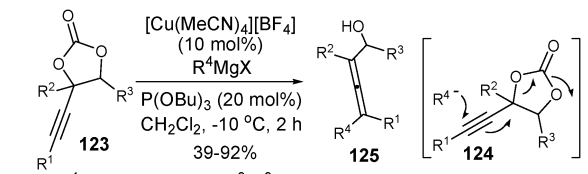
Scheme 41



Scheme 42

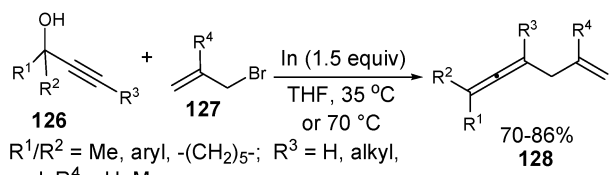


Scheme 43



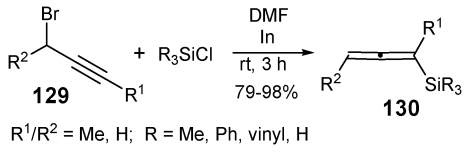
R^1 = H, Ph, Me, Et; R^2/R^3 = H, Me, -$(CH_2)_4$-; R^4 = alkyl, aryl

Scheme 44



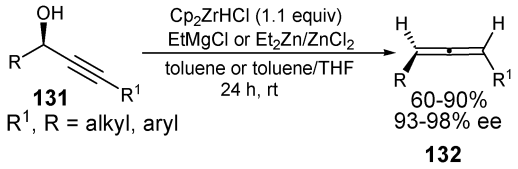
R^1/R^2 = Me, aryl, -$(CH_2)_5$-; R^3 = H, alkyl, aryl; R^4 = H, Me;

Scheme 45



Scheme 46

Allenes can be synthesized *via* the direct S_N2' addition of hydride to propargylic alcohols. Especially, aluminum hydride reagents, such as lithium aluminum hydride ($LiAlH_4$), and diisobutylaluminum hydride (DIBAL-H), have been widely used to prepare allenes from propargyl moieties such as ethers, halides, alcohols and epoxides.[102] Recently, $Cp_2Zr(H)Cl$ has been applied to the anti-S_N2'-type reductive substitution of the *in situ* generated zinc or magnesium alkoxides of propargylic alcohols **131** furnishing allenes **132** in good yields and in high optical purities (Scheme 47).[103] This approach provides direct and stereospecific access to allenes directly from propargylic alcohols and therefore greatly improved the efficiency.



Scheme 47

The rhodium-catalyzed related reactions have also been reported.[104] With the success on the synthesis of allenylboronates, Sawamura *et al.* developed a Rh-catalyzed coupling reaction between propargylic carbonates **133** and silylboronates **134** affording allenylsilanes **135** with high yields (Scheme 48).[105] The reaction of an optically active propargylic carbonate **136** took place in the presence of $[Rh(cod)_2][BF_4]$ (10 mol%) giving allenylsilanes **137** in 77% yield. From the result of $TiCl_4$-mediated addition of this allenylsilane to isobutyraldehyde, it was concluded that the center-to-axial chirality transfer in the Rh-catalyzed coupling proceeded with an *anti* stereochemistry.

Other organometallic reagents, such as organozinc, also reacted with propargylic compounds to form allenes (Scheme 49). It was found that DMSO solvent could dramatically improve the regioselectivity of the S_N2' reactions of propargyl mesylates **139** with organozinc reagents. By using LiCl-free diorganozinc, stereoselective conversion of chiral substrates **141** was also achieved.[106]

Nickel(II)-catalyzed substitution of propargylic compounds **143** with intramolecular carbon nucleophiles afforded indene derivatives **144** with an allene moiety in moderate to excellent yields (Scheme 50).[107] The reaction was initiated by nucleometallation of C≡C triple bond, followed by elimination of Ni–OR^3 bond.

Direct substitution of the hydroxy group in propargylic alcohols with nucleophiles provides an atom efficient and environmentally sound way for the introduction of alkyne or allene group into a molecule.[108] Zhou *et al.* developed a $FeCl_3{\cdot}6H_2O$ catalyzed intramolecular substitution reaction of benzylamino-substituted propargylic alcohols **145** to give tetrahydroisoquinolines **146** with *exo*-allene moiety (Scheme 51).[109] The iron catalyst helped to cleave the hydroxyl group to form an alkynyl cation intermediate.

The reaction of alkynyl oxiranes **147** with arylboronic acids **148** catalyzed by rhodium afforded *syn*-configured α-allenols **151** stereoselectively (Scheme 52).[110] The high diastereoselectivity was attributed to pre-coordination of the oxygen atom of the oxirane ring to the rhodium center. The reaction is stereospecific as the resulting *syn*-configured α-allenols showed identical ee values as compared to those of the starting oxiranes.

Alknylaziridines **152**, upon treated with 9-BBN, were converted into unprotected α-amino allenes **154** in a highly diastereoselective *syn* hydride migration (Scheme 53).[111]

2.3 By addition of propargyl or allenyl metallic intermediates to aldehydes or related compounds

Allenyl or propargylic organometallic reagents,[112] such as propargylic lithium,[113] boron,[114] silicon,[115] and tin derivatives,[116] reacted with carbonyl or related compounds,[117] including imine and oxocarbenium ion,[118] giving α-allenic alcohols or related compounds. In addition, a variety of other propargylic/allenic organometallic reagents including titanium,[119] indium,[120] aluminum,[121] zinc,[122] samarium,[123] cerium,[124] and antimony,[125] have also been found suitable for this reaction. As a special case, aldehydes interacted with alcohols in the presence of Lewis acids forming more electrophilic oxocarbenium ion species, which upon interception by nucleophilic propargyl reagents, such as propargylsilanes, afforded allenyl ether compounds.[126] In those addition reactions, protocols using tin reagents ($SnCl_2$/NaI) have the most wide applications as the reagents are easily available, cheap and have a good regioselectivity and tolerance to a variety of functional groups.

Allenyl indium reagents, generated *in situ* from indium and ethyl 4-bromobutynoate **156**, coupled with aliphatic or aromatic aldehydes **155** to afford highly functionalized α-hydroxyalkyl allenic esters **157** (Scheme 54).[127] Alcaide *et al.* have applied this protocol to the synthesis of β-lactams allenes with the ultimate goal to form functionalized pyrroles.[128]

4-Acetoxy-2-azetidinones **158** reacted with 1,6-dibromo-2,4-hexadiyne **159** in the presence of LiCl in DMF producing 2-azetidinones **160** with 1,2,4,5-hexatetraen-3-yl group on the C4-position (Scheme 55).[129]

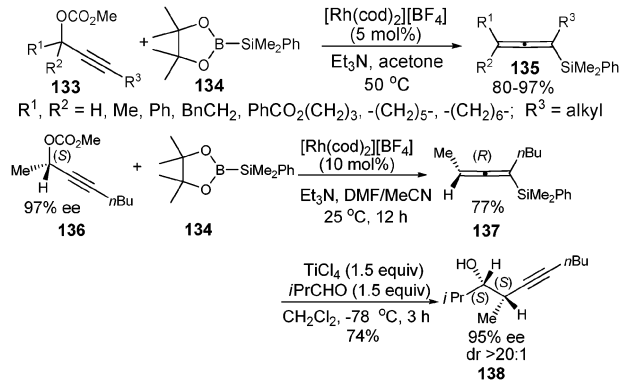
Metal-mediated reactions in aqueous media have found considerable applications in organic synthesis. Indium has a good compatibility to water. Thus, indium-mediated coupling of various electrophiles including paraformaldehyde with difluoro-substituted propargyl bromide **161** in a mixed solvent of tetrahydrofuran and water gives difluoroallene derivatives **162** and **163** (Scheme 56).[130]

Although a lot of methods for the preparation of allenic alcohols are available, only a very limited number of catalytic asymmetric allenylation have been reported.[131] Recently, the nucleophilic addition of organochromium reagents to aldehydes has shown excellent chemoselectivity, and broad functionality compatibility.[132] Several protocols based on chiral chromium catalysts have been developed for the synthesis of α-allenic alcohols.[131d] In 2007, Yamamoto *et al.* has reported an

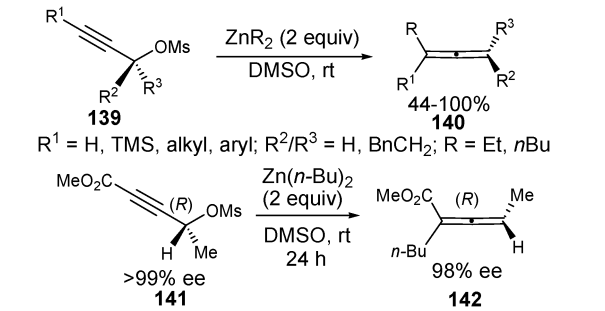
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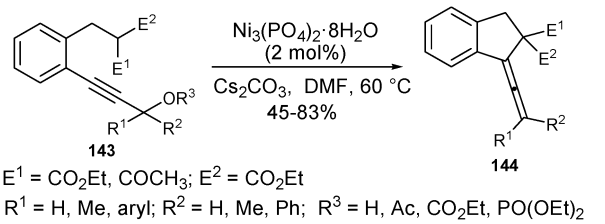
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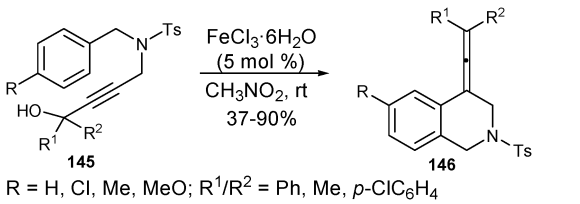
Scheme 48



Scheme 49

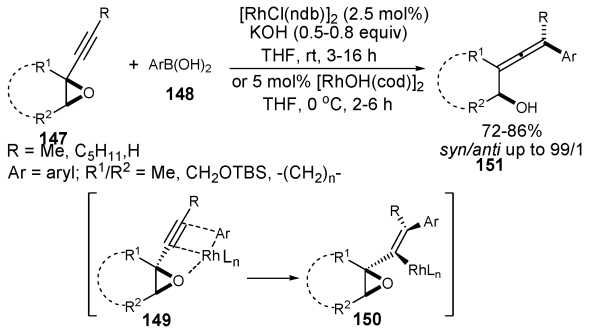


Scheme 50

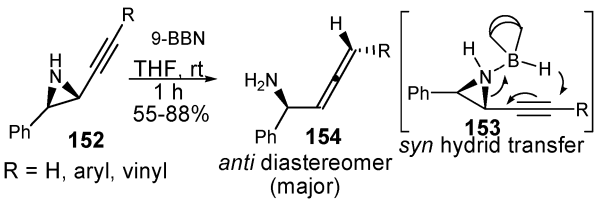


Scheme 51

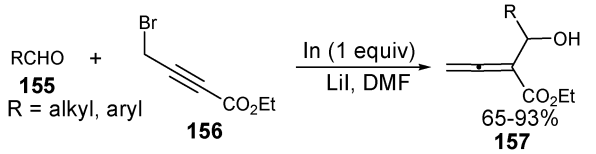
enantioselective protocol, which allows the preparation of chiral α-allenic alcohols with high enantioselectivity (up to 97% ee) and high yields (up to 91%) by directly coupling various aldehydes with 1-trimethylsilyl-3-bromo-2-butyne at room temperature using chromium complex, TBOxCr(III)Cl, developed in their group.[133] Carbazole tridentate ligand has also been applied to this allenylation with *ee* values ranging from 72–83%.[131d,134] The efficiency of this class of chiral ligand was investigated by Connell *et al.* once again. α-Allenols **167** were obtained in good yields (58–88%) and enantioselectivities (55–78% ee) (Scheme 57).[135]



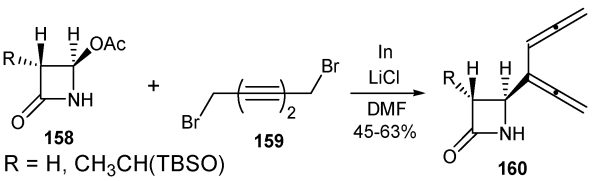
Scheme 52



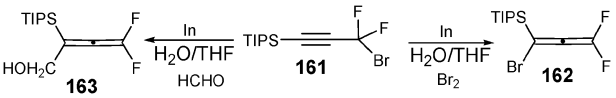
Scheme 53



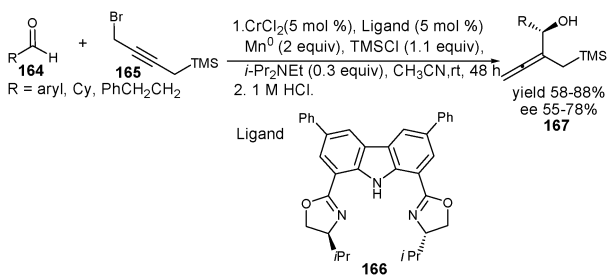
Scheme 54



Scheme 55



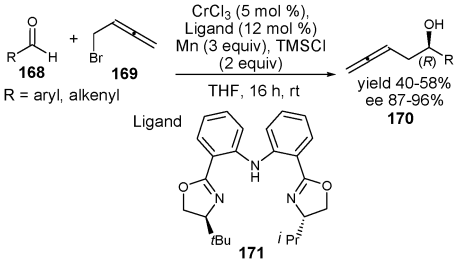
Scheme 56



Scheme 57

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Recently, Guiry *et al*. developed a Cr/Carbazole tridentate ligand catalytic system, which catalyzed the addition of bromoallene **169** with aldehydes **168** giving chiral β-allenols **170** (Scheme 58).[136] This transformation was in sharp contrast to Yamamoto's TBOxCrIIICl catalytic system, as the latter afforded 1,3-butadien-2-ylcarbinols.[137]



Scheme 58

The intramolecular addition of propargyl silane **172** to aldehyde moiety in the presence of $MeSO_3H$ provided allenols **173** as a single diastereoisomer (Scheme 59).[138] Temperature control at −78 °C was critical, otherwise enone products would be furnished.

Treatment of acetylenic ω-keto-2-alkynonates **174** with $TBSOTf/NEt_3$ formed an alkynyl enolate intermediate **175**, which was intercepted by an intramolecular ketone carbonyl group to give fused tricyclic allenoates **176** in good yields (Scheme 60).[139]

Furthermore, the alkynyl enolate intermediate could also undergo reaction with other electrophiles such as halogenating reagents.[140] It was found that when *N*-fluorobenzenesulfonimide was used, the addition took place exclusively at the α-carbon forming α-alkynyl-α-fluoro ester.[141] In contrast, the addition of the silyl ether protected alkynyl enolate intermediate **177** with I_2 gave solely the γ-product **178** (Scheme 61). The corresponding unprotected alkynyl enolate did not give the desired products which might be due to the unstable nature of the corresponding propargylic or allenylic products under the strong basic conditions.

2.4 By Crabbé homologation reaction of terminal alkynes and related reactions

Crabbé homologation reaction produces monosubstituted allenes from readily-available terminal alkyne, formaldehyde, iPr_2NH and CuBr in dioxane (Scheme 62). This reaction involves copper-catalyzed addition of terminal alkyne to the *in situ* formed Schiff base and hydride migration to the C≡C bond in an S_N2' substitution manner.[142]

The useful of this reaction has inspired several modifications to improve efficiency or expand the reaction scope. It was found that microwave irradiation could accelerate reaction rate, reducing reaction time from usually 6–24 h to 1–20 min (Scheme 63). Bisallenes were also efficiently prepared from bisalkynes under this condition.[143]

The replacement of diisopropylamine with dicyclohexylamine and CuBr with CuI improved the reaction yields and expanded the substrate scope (Scheme 64).[144] Many functional groups such as mesylate, hydroxyl group, ether, amide, *etc*. were tolerated under the current conditions.

Crabbé homologation reaction suffers from a serious limitation; it works only for the preparation of monosubstituted allenes. Recently, this group developed an ZnI_2-mediated protocol for synthesis of 1,3-disubstituted allenes **189** from 1-alkynes **186**, aldehydes **187**, and morpholine **188** (Scheme 65).[145] Allenols or allenyl amines may be prepared from terminal propargyl alcohols or amines. However, this protocol is still not suitable for the preparation of trisubstituted allenes from ketones.

Based on early mechanistic studies for this reaction, propargylamine derivatives were explored directly as the starting materials for the synthesis of allenes. This stepwise strategy resolved parts of problems that Crabbé homologation has, showing merits such as shorter reaction time, lower temperature and catalyst loading. In addition, di- or trisubstituted allenes are also accessible. Many transition metals are found to be able to catalyze the reaction.

Nakamura *et al*. developed a catalytic system that utilized palladium with electron-deficient phosphine ligands $(C_6F_5)_3P$ as the catalyst (Scheme 66).[146] Mono-substituted allenes **191** were synthesized from the corresponding propargylamines **190** with up to 99% yield, whereas 1,3-disubstituted allenes were obtained with much lower yields (36–68%).

Asymmetric variants also emerged by utilization of the easily available chiral amines. Che *et al*. reported that chiral propargylamines **192** afforded axially chiral allenes **193** under the catalyst of $KAuCl_4$ in CH_3CN at 40 °C (Scheme 67).[147] High yields (up to 93% yield) and excellent enantioselectivities (up to 97% ee) were realized under those conditions.

Furthermore, a silver(I)-mediated variant was developed by the same group, which showed compatible results as compared with the much expensive gold catalyst (Scheme 68).[148] The use of microwave irradiation again allowed the completion of the reaction in a much shorter time than that required under conventional thermal conditions.

Propargyl benzyl ether derivatives **196** could also serve as starting compounds for the synthesis of variously substituted allenes **197** under the action of a gold(I) catalyst (Scheme 69).[149] The reaction proceeded *via* a 1,5-hydride transfer/elimination of benzaldehyde sequence.

2.5 By palladium-catalyzed cross-coupling reactions

Palladium-catalyzed cross-coupling reactions of propargyl substrates with organometallic reagents or nucleophiles constitute an important method to allenic compounds.[150] For Pd(II)-mediated reactions, the initial step usually involved nucleopalladation (Scheme 70).[1j,k] For Pd(0)-mediated coupling, the reaction may be initiated by oxidative addition of Pd(0) to the corresponding propargylic compounds **2-VIII** with a good leaving group forming allenylpalladium intermediate **2-VIV**[151] or oxidative addition of Pd(0) to alkenyl (aryl) halide **2-VVVI**,[152] followed by transmetallation with propargyl/allenyl metals **2-VVVIII** and **2-VVVIV** (Scheme 71 and Scheme 72).[153]

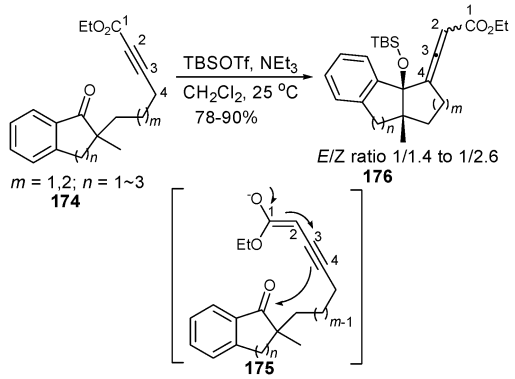
The allenylpalladium intermediate formed by oxidative addition of Pd(0) to propargylic carboxylate could be trapped

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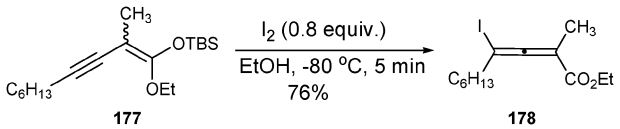
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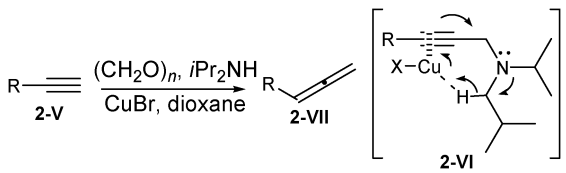
Scheme 59



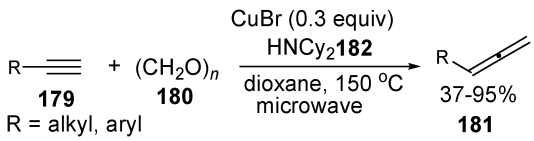
Scheme 60



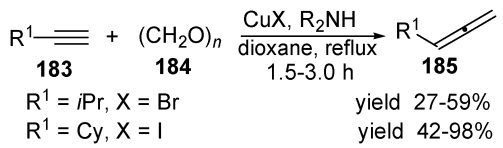
Scheme 61



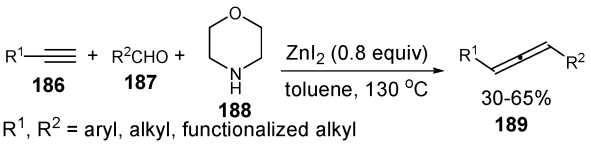
Scheme 62



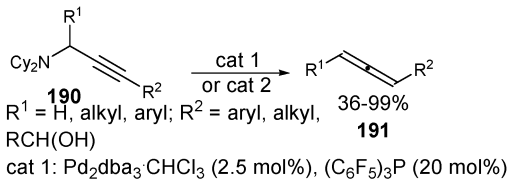
Scheme 63



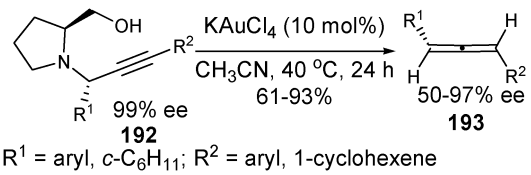
Scheme 64



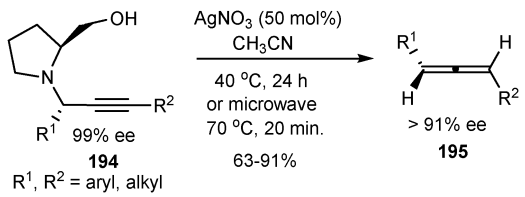
Scheme 65



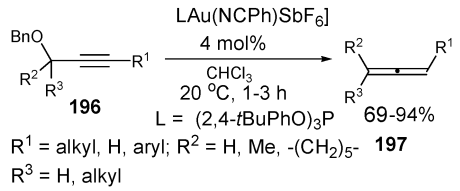
Scheme 66



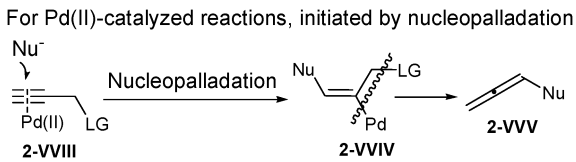
Scheme 67



Scheme 68



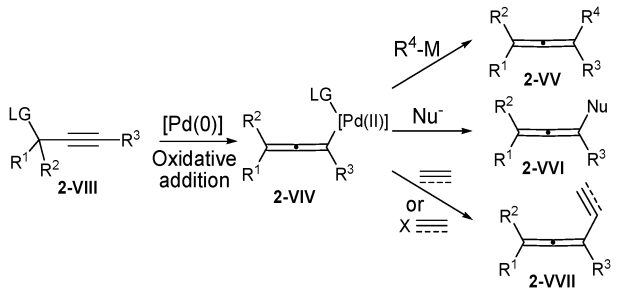
Scheme 69



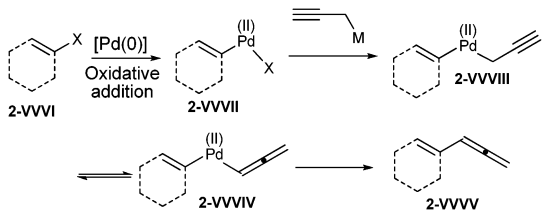
Scheme 70

by nucleophiles, which gave a variety of functionalized allenes.[154] Liang *et al.* intercepted the allenylpalladium complex by carbon nucleophiles generated *in situ* by Michael addition of arylamines **199** to diethyl 2-{2-[3-(ethoxycarbonyloxy)prop-1-ynyl]benzylidene} malonate **198** (Scheme 73).[155] This tandem reaction offered an efficient route to highly substituted indenes with an allene moiety **202**. Ni(II) also worked well for this reaction, which coordinated to the triple carbon–carbon bond to facilitate the attack of nucleophiles.

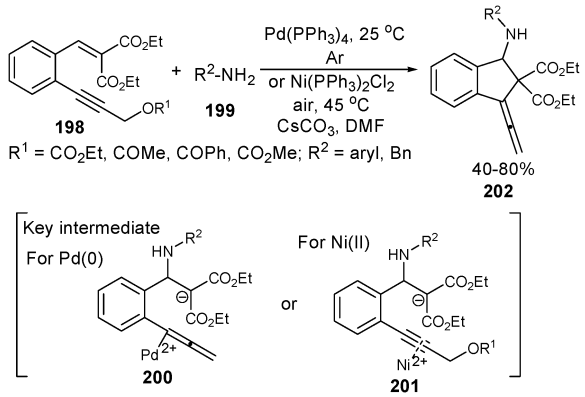
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Scheme 71

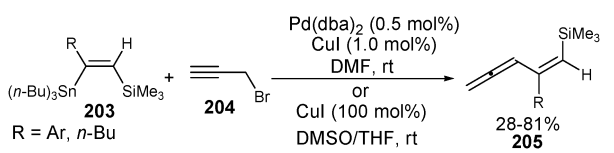


Scheme 72



Scheme 73

Conjugated 1,3,4-trienyl silanes **205** were prepared by a combined $Pd(dba)_2$/CuI catalyst or copper(I) iodide-mediated reaction of silyl(stannyl)ethenes **203** with propargyl bromide **204** (Scheme 74).[156] In most cases, the CuI-mediated catalytic system afforded better yields than the corresponding palladium catalysis. The configuration of all products was *E*, except for butyl substituted substrates. Organozirconium reagents, such as α-phosphonozirconacyclopentenes or α-borylzirconacyclopentenes, may also undergo similar reactions.[157]



Scheme 74

Treatment of propargylic ether **206** with $Cp_2Zr(n\text{-}Bu)_2$ led to allenic/propargylic zirconium reagents (Scheme 75). The ratio depended on the substitution pattern of starting propargylic ethers. Aryl- or alkyl-substituted propargylic ethers favored the allenic zirconium, whereas TMS-substituted propargylic ethers preferred to produce alkyne zirconium. These reagents, upon coupling with aryl iodides in the presence of $Pd(PPh_3)_4$/CuCl, gave multi-substituted allenes **210** in moderate to good yields.[158]

In 2006, we found that allenylpalladium complex could be inserted into a functionalized allene moiety to afford heterocycle substituted allenes.[159] Recently, we expanded this kind of reaction to palladium(0)-catalyzed coupling-cyclization reaction of propargylic carbonates **211** with 2-(2′,3′-allenyl)acetylacetates **212** (Scheme 76). The reaction proceeded smoothly under neutral conditions, affording the *O*-attacked five-membered products 1,3,4-alkatrien-2-yldihydrofurans **213** exclusively in good to excellent yields. The products were further manipulated to the synthesis of polysubstituted 2-(dihydrofuryl) cyclopentenone derivatives **214** *via* a catalytic Pauson–Khand reaction under ambient conditions.[160a]

However, when (2,3-butadienyl)malonates **216** or bis(phenylsulfonyl)methanes were subjected to similar reaction conditions, three-membered ring products, polysubstituted 1,3,4-alkatrien-2-yl cyclopropane derivatives **217** were afforded in 73–96% yields (Scheme 77).[160b]

A Pd(II)-mediated cross-coupling reaction of alkynoates **218** and propargyl alcohols **219** to conjugate alkenyl allenes **220** were accomplished by Jiang *et al.* (Scheme 78).[161] Chloropalladation of alkyne initiated the reaction, followed by insertion of the carbon–carbon triple bond of the propargylic alcohol into the carbon–palladium bond of the vinylpalladium species and subsequent β-OH elimination finished the catalytic cycle.

In the past few years, we have studied the Negishi coupling reactions involving propargyl/1,2-allenyl metal species to afford allenes or alkynes in a highly selective manner. We found a dramatic dependence of regioselectivity on the steric factors and electronic of substrates.[162] We also applied 1,3-H/Li shift[163] to the Pd-catalyzed Negishi cross-coupling reaction of 3-iodoalk-2-enoates **222** with the propargyl/1,2-allenyl zinc species giving conjugated alkenyl allenes *E*-**223** and *E*-**225** (Scheme 79).[164] A remarkable ligand effect on the regioselectivity and reactivity was also observed.[165]

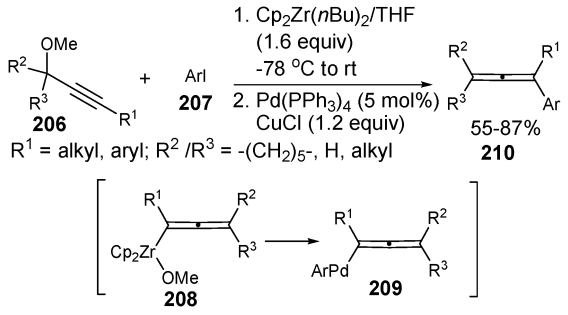
The cross-coupling reactions of 3-tri(*n*-butylstannyl)-1-(trimethylsilyl)-1-propyne **228** with aryl or alkenyl iodides **227** yielded conjugated 1,1-disubstituted allenylsilanes **231** (Scheme 80).[166] The proposed mechanism involved an isomerization process of alkynyl palladium intermediate **229** to allenyl palladium intermediates **230**.

Oshima *et al.* have reported a palladium-catalyzed reaction of homopropargylic alcohols **232** with various aryl bromides **233** forming di-, tri- or tetrasubstituted arylallenes **236** (Scheme 81). Diastereoselective synthesis of tetrasubstituted allenes was also feasible. It was proposed that the reaction involved a σ-allenylpalladium intermediate **235** from retro-propargylation of the corresponding homopropargyl alcohols.[167]

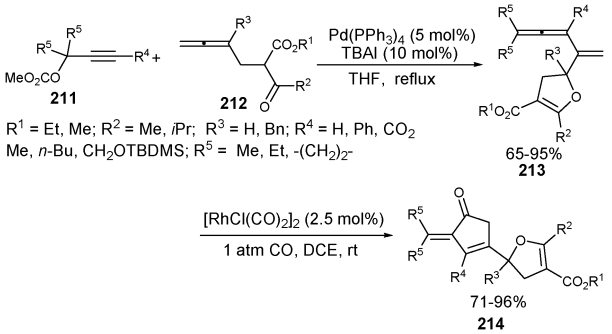
Palladium-catalyzed decarboxylative hydrogenolysis of propargylic alcohol derivatives provided a useful method for synthesizing allenes or alkynes.[168] The regioselectivity was found to be highly dependent on the structure of the

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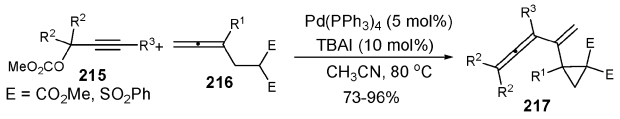
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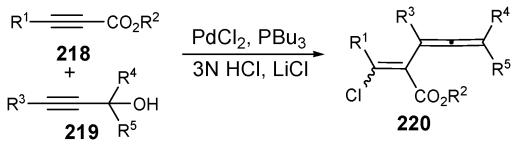
Scheme 75



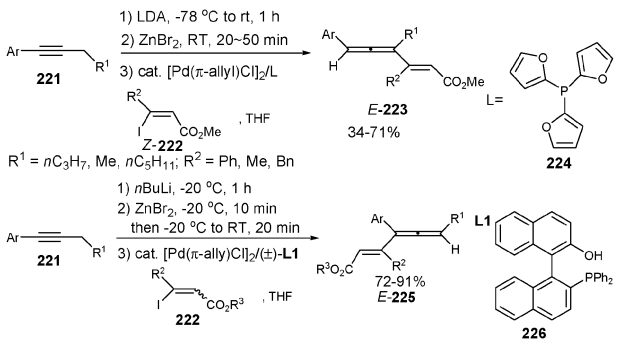
Scheme 76



Scheme 77

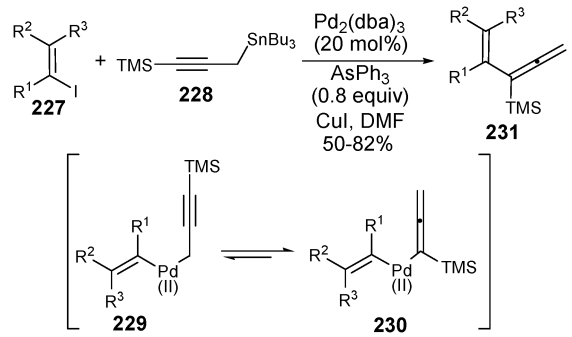


Scheme 78

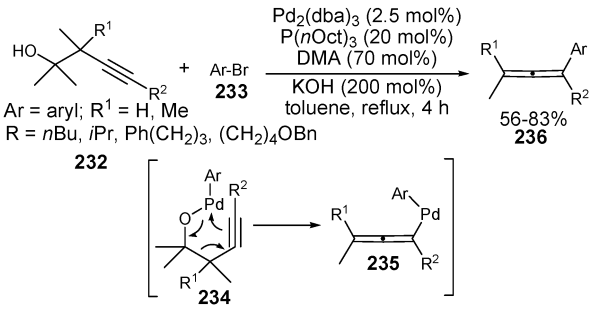


Scheme 79

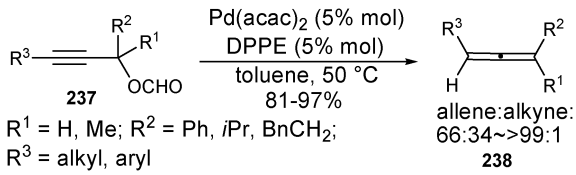
substrates. Most recently, Sawamura *et al*. developed a ligand-controlled regioselective palladium-catalyzed decarboxylative hydrogenolysis of propargylic formates **237**, which allowed selective synthesis of allenes or alkynes with a significantly expanded substrate scope.[169] When DPPE was used as ligand, internal allenes **238** were obtained with excellent selectivity



Scheme 80



Scheme 81

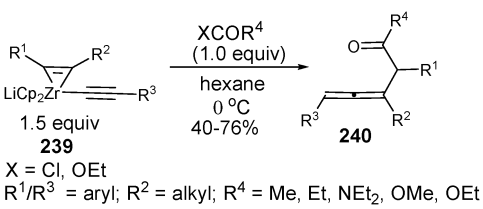


Scheme 82

(Scheme 82). Changing the ligand from DPPE to DPPH resulted in α-selective hydrogenolysis, which formed the corresponding alkynes.

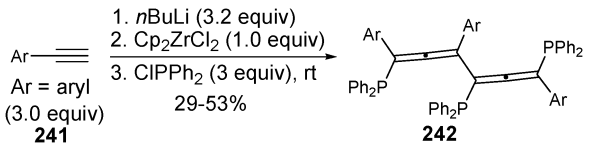
2.6 By zirconocene-mediated reactions

Alkynylzirconate complexes **239**, formed *in situ* from terminal alkynes, alkyllithium reagents, and zirconocene dichloride, reacted with carbonyl compounds, such as ethyl chloroformate, ethyl acetate, and diethylcarbamic chloride, afforded highly substituted 2,3-allenyl ketones **240** (Scheme 83).[170]

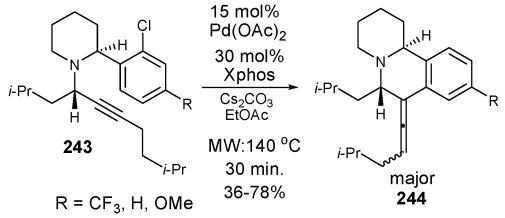


Scheme 83

Conjugated bis(allene)s **242** could be obtained from zirconium-promoted linear coupling of alkynes **241** in high yields (Scheme 84).[171]



Scheme 84



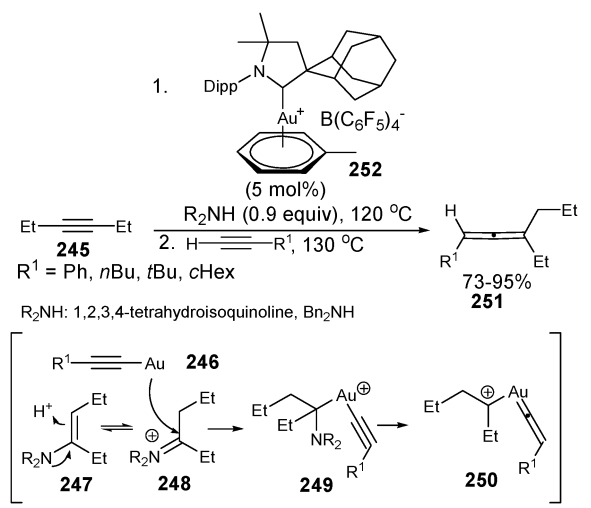
Scheme 85

2.7 Miscellaneous reactions

It has been known that aryl bromides, especially *ortho*-substituted ones, react with dialkylacetylenes in the presence of a palladium catalyst to give the corresponding aryl-substituted allenes.[172] Recently, Hamblett *et al.* prepared tricyclic hexahydro-2*H*-pyrido[2,1-a]isoquinoline allene **244** *via* an intramolecular Heck cyclization of an aryl chloride with an alkyne **243** (Scheme 85).[173] Both ligands and reaction solvents exerted effects on the regioselectivity of the allene.

Bertrand *et al.* have developed a protocol for synthesis of multi-substituted allenes **251** by a cationic gold complex catalyzed coupling of alkynes **245** (Scheme 86).[174] The gold complex played two roles: promoting the hydroamination of internal alkynes[175] and catalyzing addition of terminal alkynes to the resulted enamines **247**. The latter process might involve an Au(carbene)(vinylidene) intermediate and amine acting as a two hydrogen donor. When unsymmetrical internal alkynes were used, two isomeric allenes, arising from the Markovnikov and anti-Markovnikov hydroamination products could be obtained in a somewhat low regioselectivity.

Gold-catalyzed intramolecular reactions of 3-alkynyl-bearing indoles **253** led to highly functionalized 3-allenylindoles **255** (Scheme 87).[176] The reactions involved a sequence including cyclization/heterolytic fragmentation/elimination steps.



Scheme 86

Treatment of propiolate **257** and aromatic 1,2-diketone **256** with a catalytic amount of base produced a novel ambiphilic allene **258** (Scheme 88).[177] However, this allene was so reactive, which dimerized to the corresponding cyclobutane **259** when no external chemical intercepting reagent was presented.

Ruthenium complex combined with an acidic additive catalyzed the formation of allenyl carbamates **263** from propargyl alcohols **260** and isocyanates **261** (Scheme 89).[178] The yields were not satisfactory due to the formation of symmetrical urea derivatives $R^2NHCONHR^2$ and recovery of starting material.

3. Allenes from conjugated enynes

Conjugated enynes, which are readily available by the Sonogashira coupling reaction, have been widely used for the synthesis of multi-substituted allenes.

3.1 By 1,4-addition of conjugated enynes

1,4-Addition of substrates to conjugated enynes represented an efficient synthetic tool for the generation of multi-substituted functionalized allenes (Scheme 90).[58c,179] A variety of substrates, such as lithium aluminum hydride, hydroboranes, hydrosilanes, hydrogen halides, halogens, organometallic reagents, carbon pronucleophiles, and acyl chlorides have been used.

3.1.1 Intermolecular 1,4-addition. Carbolithiation[180] of conjugated enynes leads to the formation of allenyllithium intermediate **3-IV** and **3-V** (Scheme 91),[102,103] which can be trapped by electrophiles,[163d,181] or participate in palladium-catalyzed coupling reactions.[163d,182] Commercially available alkyl lithium, such as *n*-BuLi, *sec*-BuLi, and *tert*-BuLi are often used as starting organolithium compounds. The use of aryllithiums is less developed due to slow carbolithiation rate at low temperature and side reactions.

Recently, Yoshida *et al.* partly solved this problem by carrying out the carbolithiation process in an integrated microflow system consisting of three T-shaped micromixers and three microtube reactors.[183] Enynes **265** reacted with aryllithiums and chlorosilanes in this apparatus furnished allenylsilanes **266** in good yields and high regioselectivity (Scheme 92).[184] Precise residence time control was responsible for the success of the transformation.

1,4-Addition of "frustrated Lewis pairs" (pairs of strong Lewis acids and bases bearing bulky substituents)[185] $Mes_2PCH_2CH_2B(C_6F_5)_2$ **267** to conjugated enynes **268** formed eight-membered cyclic allenes **269** and **270** (Scheme 93).[186]

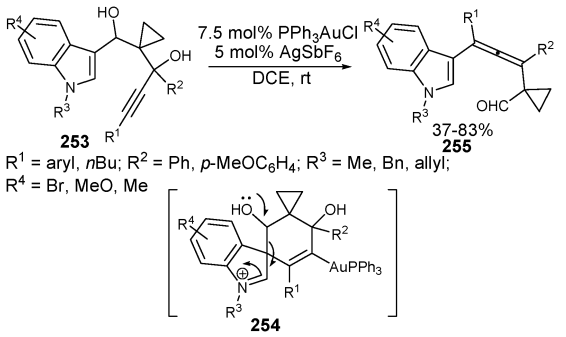
Zhang *et al.* have shown that electron-deficient 1,3-conjugated enynes **271** underwent base-catalyzed Michael addition giving highly functionalized 1,2-allenes **273** (Scheme 94).[187] The regioselectivity was found to be controlled by the nature of nucleophiles. If heteroatom nucleophiles were applied, 4*H*-pyrans were afforded by a formal [3 + 3] cycloaddition.

Later, the above reaction was expanded to Pd-catalyzed three-component tandem Michael addition/cross-coupling reactions (Scheme 95).[188] Tetrasubstituted allenes **277** were obtained from electron-deficient enynes **274** with nucleophiles **276** and aryl halides **275** in moderate to good yields.

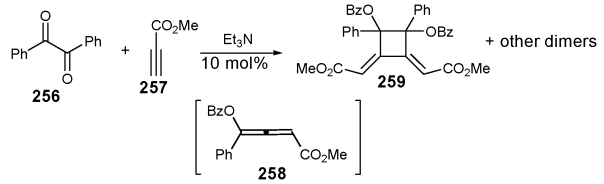
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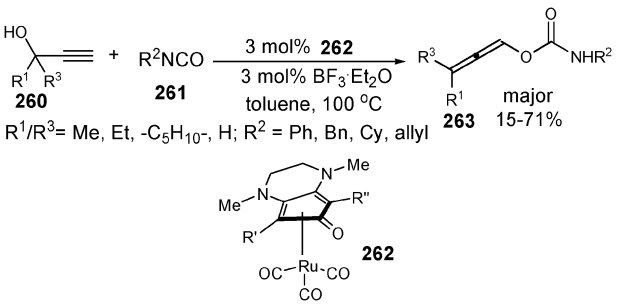
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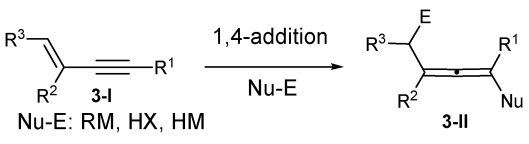
Scheme 87



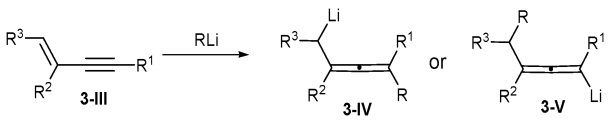
Scheme 88



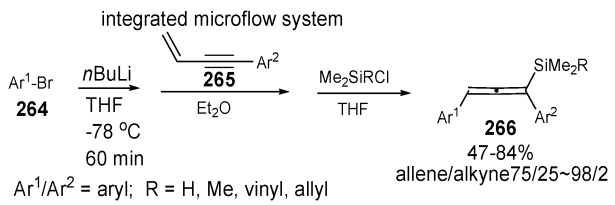
Scheme 89



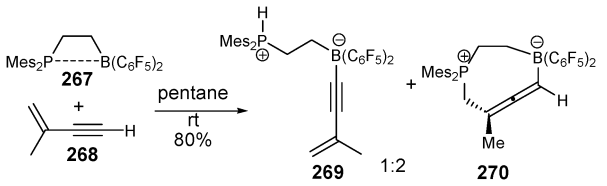
Scheme 90



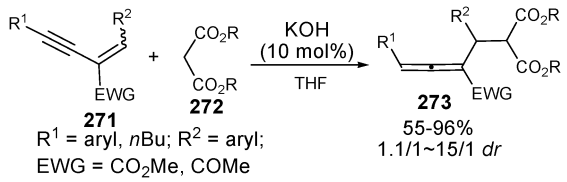
Scheme 91



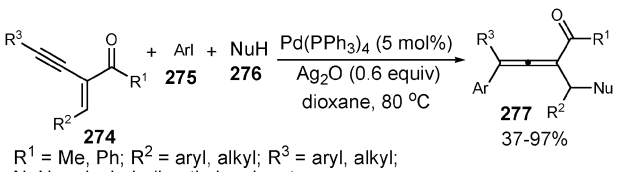
Scheme 92



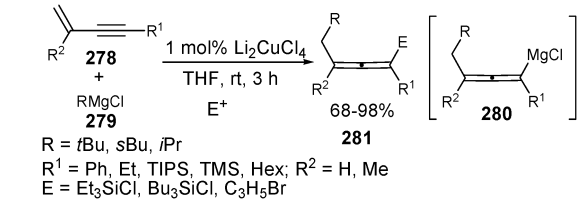
Scheme 93



Scheme 94



Scheme 95

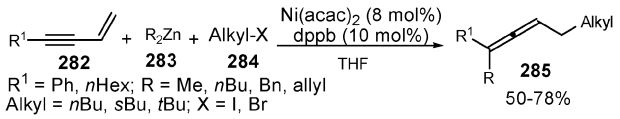


Scheme 96

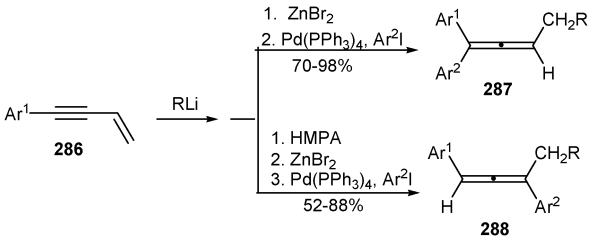
Terao and Kambe *et al.* developed a copper-catalyzed carbomagnesiation of enynes **278** with *sec*- and *tert*-alkyl Grignard reagents **279** to give allenyl Grignard reagents **280**, which could be trapped with a variety of electrophiles such as alkyl and allyl bromides, chlorosilanes, enones, benzoyl chlorides, and carbon dioxide (Scheme 96).[189]

Furthermore, a nickel-catalyzed three-component reaction of alkyl halides **284**, enynes **282** with organozinc reagents **283** is developed by the same authors (Scheme 97).[190] The reaction was proposed to proceed *via* nickelate complex initiated the formation of alkyl radical species by single electron transfer, and the addition of alkyl radicals to unsaturated C–C bonds, followed by reductive elimination of Ni(II) intermediates.

A protocol was established by us to control the 1,3-lithium shift in propargylic/allenylic species formed by the conjugate addition of 4-aryl-3-butyn-1-enes **286** with organolithiums. It was found that the addition of HMPA determined the



Scheme 97



Scheme 98

regioselectivities of the formed organolithium species. Subsequent transmetalation and Pd(0)-catalyzed cross-coupling with aryl halides afforded 1,1-diarylallenes **287** and 1,3-diarylallenes **288**, respectively (Scheme 98).[163d]

This group has demonstrated that the direct carbomagnesiation of 5-perfluoroalkylpenta-4(*E*)-en-2-ynols **289**, followed by *anti*-defluorometallation gave 6-perfluoroalkyl-6-fluoro-2,3,5(*Z*)-alkatrienols **291** in a *Z*-selective manner (Scheme 99).[191]

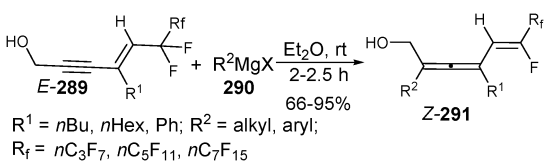
1,4-Silaboration of conjugate enynes **292** catalyzed by Pt-complexes afforded allenes **294** bearing boronic and silyl substituents (Scheme 100).[192] This addition was limited to substrates with bulky substituents in the alkynyl moiety and no substituents in the alkenyl part.

Axially chiral allenylsilanes **298** were prepared highly selectively by rhodium/chiral diene complex catalyzed asymmetric addition of arylboronic acids **296** to enynamides **295** (Scheme 101).[193]

3.1.2 Intramolecular 1,4-addition. Tang *et al*. reported the intramolecular hydroamination of enynyl amine **299** to yield allenyl substituted five-membered pyrrolidines **300** in the presence of a catalytic amount of *n*-BuLi with up to 95% yield and 5 : 1 diastereomeric ratio. *trans*-Enynes are essential for the successful cyclization (Scheme 102).[194]

Enantioselective halocyclization reactions provide an efficient and versatile tool for the synthesis of chiral cyclic compounds.[195] Braddock *et al*. studied the stereochemical course of bromoetherification of enynes by X-ray crystallographic analysis and by stereospecific synthesis of authentic bromoallenes, which showed that intramolecular bromoetherification of enynes were highly stereospecific and proceeded *via* a stereoselective *syn*-addition.[196] Enynyl acids **301** underwent *syn*-1,4-bromolactonization in the presence of DABCO and NBS to give lactones **302** tethered with multi-substituted allenes under mild conditions (Scheme 103). Both (*Z*)- and (*E*)-enynes worked well and yielded axially chiral allenes with a stereogenic center at the α-position.[197]

An asymmetric variant was further explored. Cinchonidine derived bifunctional chiral reagent efficiently catalyzed *syn*-1,4-bromolactonization of conjugated (*Z*)-enynyl acid **303** (Scheme 104).[198] Diastereomeric ratios >20 : 1 and up to 99% enantiomeric excesses were observed.



Scheme 99

3.2 Cyclometallation of enynes

The cyclometallation of an alkyl-substituted conjugated enyne **307** with a zirconocene-ethylene complex afforded five-membered zirconacyclopentene **308** regioselectively with the alkenyl substituent located on the α-position of a zirconacycle (Scheme 105).[199] The interaction of the alkenyl moiety with the zirconium atom might contribute to the observed high regioselectivity. When an aldehyde was added to this intermediate, β-hydroxyallene **311** with multiple stereogenic centers was furnished in moderate yield after hydrolysis. The C–C bond formation occurred selectively at the alkenyl–zirconium bond. For *E*-enynes, a high diastereoselectivity up to 98/2 was observed.

3.3 Skeletal rearrangement of 1, *n*-(hetero)enynes

3.3.1 [3,3]-Claisen-Cope rearrangement. [3,3]-Claisen-Cope rearrangement is a method of choice for preparation of β-allenic aldehydes, ketones, esters, imines and amides starting from propargyl vinyl ethers or related derivatives (Scheme 106).[200] The rearrangement under the interaction of transition metals allows for mild reaction conditions, high efficiency and wide functional group tolerance.

Vinyl propargyl ethers **312** tied with indolinone underwent the thermal Claisen transformation affording highly functionalized allene-appended oxindoles **313** in moderate to good yields (Scheme 107).[201]

The *ortho*-ester Claisen rearrangement[202] of chiral 3-halo propargyl alcohols **314** with triethyl orthoacetate in the presence of a catalytic amount of propionic acid gave enantiomerically enriched allenyl halides **316** (Scheme 108).[203] The high enantioselectivity (>93/7) is attributed to the formation of a boat conformation.

The chelating-controlled Claisen rearrangement of α-acyloxy-α-alkynylsilanes **317** afforded a useful route to optically active α-(allenyl)glycines **319** (Scheme 109).[204] The observed high diastereoselectivity (*dr* = >20 : 1) was ascribed to the preferential formation of the *Z*-enolate and subsequent rearrangement *via* a boat transition state, in which the sterically bulky silyl group was oriented in an equatorial position. The sterically bulky silyl group could be easily removed by TBAF.

Prabhakar and Lobo *et al*. reported that *N*-oxy-3-aza Cope rearrangements of **320** led to oxime allenes **321**. It was found that the *Z*-configuration of the enaminic double bond was a critical factor for the rearrangement to take place (Scheme 110).[205]

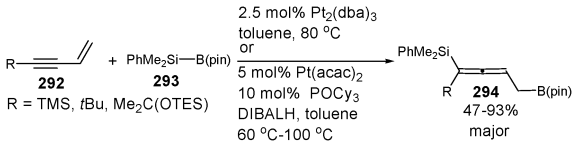
Cationic Rh(I)-dppf complex catalyzed olefin isomerizations/[3,3]-Claisen rearrangement of allyl propargyl ethers **322** to allenic aldehydes were developed by Tanaka *et al*. (Scheme 111).[206] Due to the instability of allenic aldehydes, $NaBH_4$ was added to give the corresponding reduced alcohols **323**. Temperature has a great effect on the chemoselectivity. Allenic aldehydes were furnished in good yields at room temperature, whereas at 80 °C, 4,5-dienals were obtained in good yields.

Treatment of propargyl ether **324** bearing an amide group with Tf_2O and collidine afforded α-allenyl lactones **328** in moderate yields (Scheme 112).[207] The reaction proceeded *via* nucleophilic attacking initially forming keteniminium **325** by

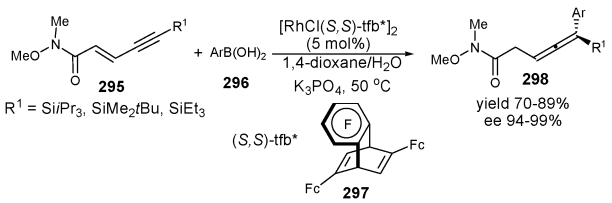
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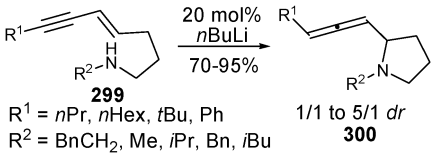
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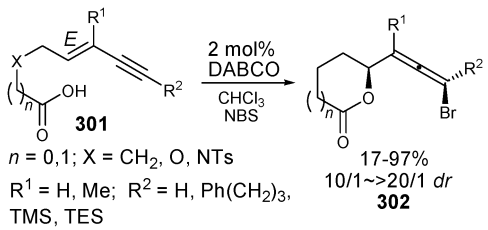
Scheme 100



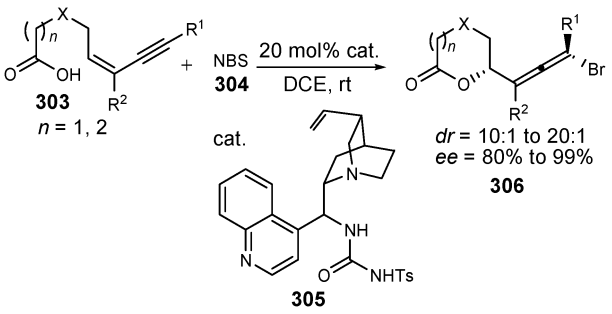
Scheme 101



Scheme 102



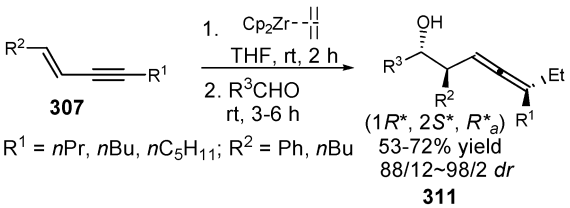
Scheme 103

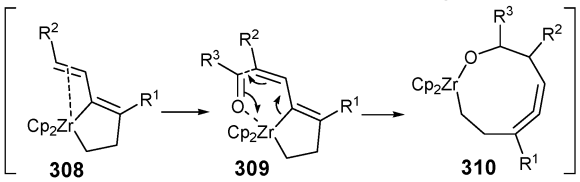


Scheme 104

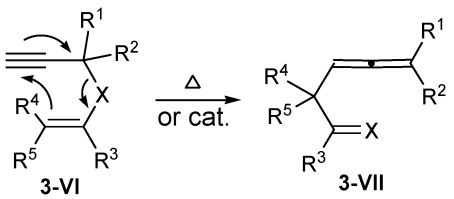
oxygen atom at ether moiety, followed by [3,3]-sigmatropic rearrangement and hydrolysis.

Allenamides **330** were formed by palladium-catalyzed [3,3]-rearrangement of propargylic phosphorimidate **329** (Scheme 113).[208] By using an enantiomerically enriched propargylic alcohol, this strategy represents a facile entry into stereochemically defined allenamides. Propargyl acetates undergoing 1,3-acyl migration to access allenes in the presence platinum(II), gold(I) and gold catalysis also underwent in a similar pathway.[209]

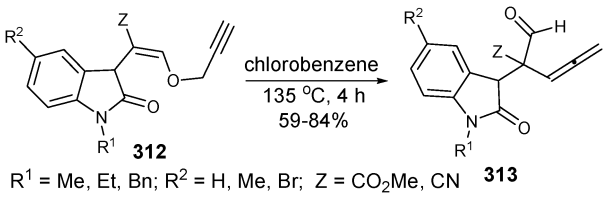




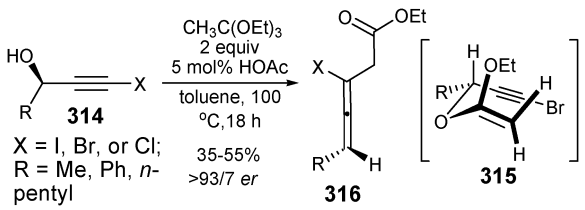
Scheme 105



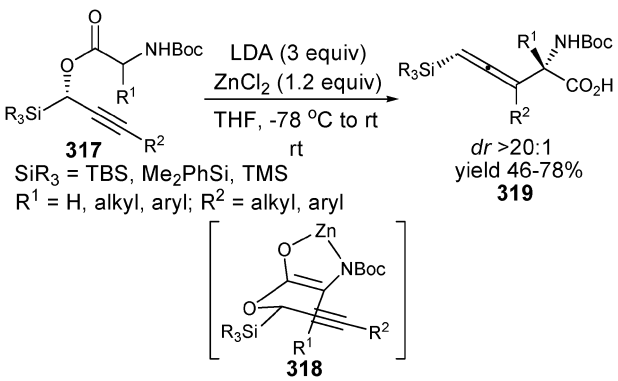
Scheme 106



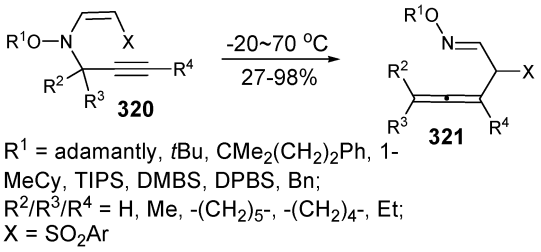
Scheme 107



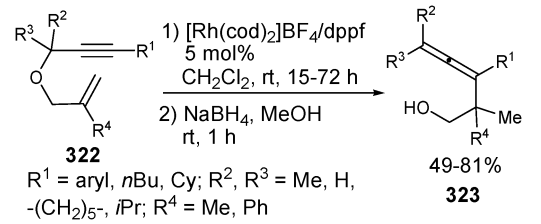
Scheme 108



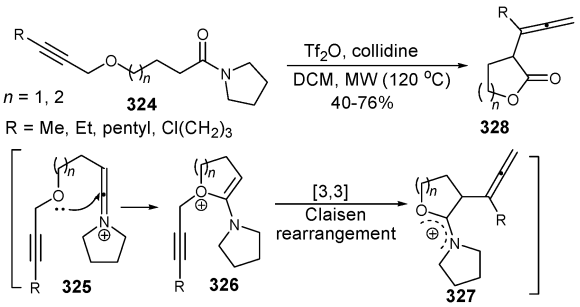
Scheme 109



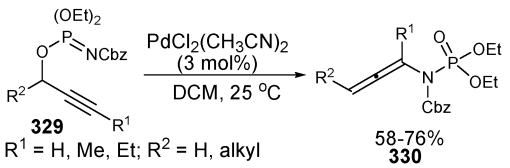
Scheme 110



Scheme 111



Scheme 112



Scheme 113

Gold complexes are known to promote 3,3-sigmatropic shift of propargyl acetate to furnish the allenyl acetate.[209] When one more alkyne moiety was introduced, the high alkynophilic gold induced a second oxacyclization/1,3-dioxolium ion generation *via* internal nucleophilic attack by the carbonyl oxygen of the acetoxy group at C-2 of allene **332** (Scheme 114).[210] Water addition and protodemetalation afforded α-allenones **335** in moderate to good yields. Controlling the reaction time to less than 15 min is critical for the allene synthesis as Au(I) further activates the allene functionality inducing the sequential 1,3-dioxole ring opening and 5-*endo*-dig carbocyclization to furnish the cyclopentenone derivatives.

3.3.2 [2,3]-Sigmatropic rearrangement. [2,3]-Sigmatropic rearrangements (Scheme 115) are often used for the synthesis of heteroatom substituted allenes, such as allenic phosphinates,[211] phosphonates,[212] sulfoxides,[2c,213] and sulfones.[214,215] The highly concerted bond breaking and bond formation pathway makes the selective center-to-axis chirality transfer possible.

The condensation of phosphite chloride **337** with propargyl alcohol **336** led to intermediate alkynyl phosphinites **338**, which underwent [2,3]-rearrangement to form allenes **339** *via* intramolecular addition of the phosphorus atom to the triple bond (Scheme 116).[216]

2-Allene-substituted 1,3,2-oxazaphospholanes **343** were obtained in 75–84% yields involving the formation acetylenic phosphite **342** from propargyl alcohol **341** with (*d*)-pseudoephedrine **340** followed by [2,3]-sigmatropic rearrangement (Scheme 117).[217]

Chiral allenamides **347** were prepared with high levels of enantiomeric purity by amination/[2,3]-sigmatropic rearrangement of enantioenriched propargylic sulfimides **344**. This reaction proceeded with a high level of chirality transfer (Scheme 118).[218]

It is well known that diazo compounds react with a sulfide to generate sulfonium ylides in the presence of transition metals. With a double asymmetric induction by combining a chiral camphor sultam auxiliary and Cu(I) catalyst with chiral ligands **350**, [2,3]-sigmatropic rearrangement of propargyl sulfonium ylides gave enantioenriched 3,4-allenols **351** in good to excellent yields (Scheme 119).[219]

Allenic sulfones are widely used in organic synthesis as partners for cyclization.[214a,b,d,220] Thermal[214i] or transition-metal catalyzed[214e,g] rearrangements of propargylic sulfinates, generated by the reaction of sulfinyl chlorides with propargylic alcohols, are the typical procedures for the synthesis of allenic sulfones. Recently, Harmata *et al.* have used $AgSbF_6$ as a catalyst to produce allenic sulfones. The reaction of various substituted propargylic sulfinates **352** with 2 mol% of this catalyst in CH_3NO_2 resulted in the rapid formation (2 min) of allenic sulfones **353** in essentially quantitative yields (Scheme 120).[221]

3.3.3 Transition-metal catalyzed cycloisomerization. A silver-mediated transformation of allyl-tethered propargyl alcohol **354** afforded vinylidenecyclopentanes **355** in moderate to good yields. Trace amounts of water was necessary for the reaction to take place (Scheme 121).[222]

Gold-promoted *5-exo-dig* cyclization of 1,6-enynes **357** giving vinylidenecyclopentanes **360** was also developed. However, other side reactions made this method less predictable (Scheme 122).[223]

4. Allenes from cyclopropane or its derivatives

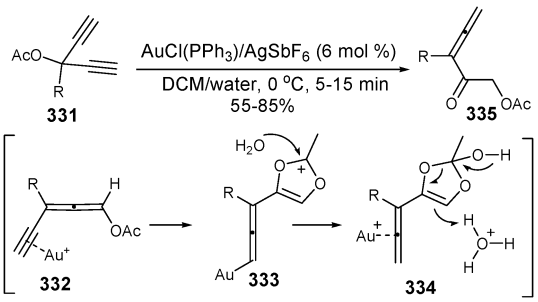
Conversion of geminal dihalocyclopropanes **4-I** to allenes **4-III** with alkyllithium reagents at low temperature (−78 °C) involved a carbine-like intermediate **4-II**. This so-called Doering–Moore–Skatteböl (DMS) reaction,[224] is less applied now compared with other methods (Scheme 123).[2a,225]

Satoh's researches revealed that sulfoxide could exchange itself with metal as halogen has done.[226] The application of sulfoxide-metal exchange in the DMS reaction afforded a new method for synthesis of allenes (Scheme 124). For example, when 1-(*p*-tolylsulfoxide)-1-chlorocyclopropropane **362** was

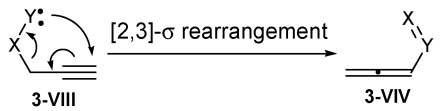
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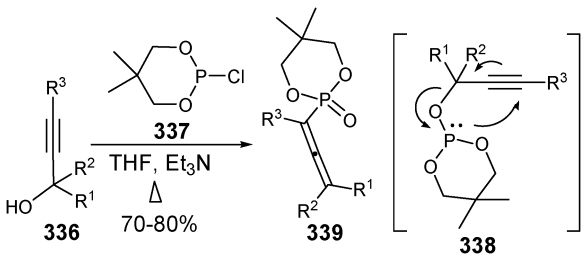
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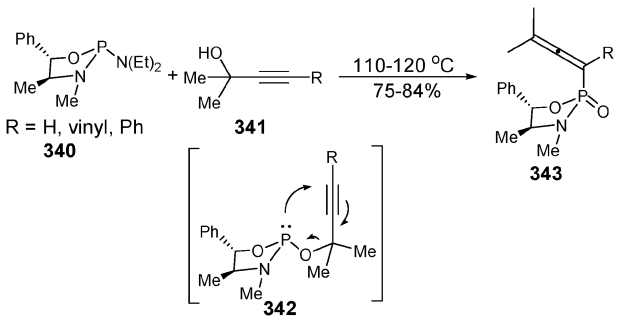
Scheme 114



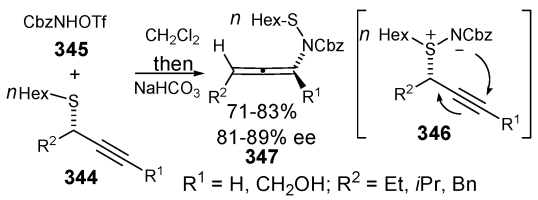
Scheme 115



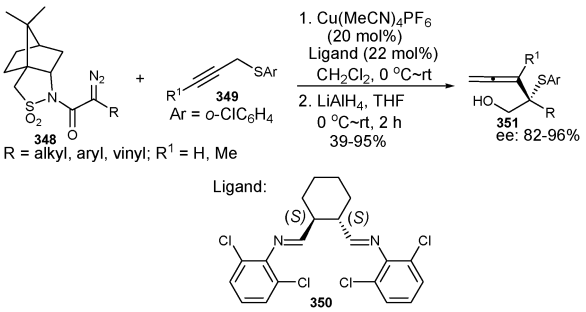
Scheme 116



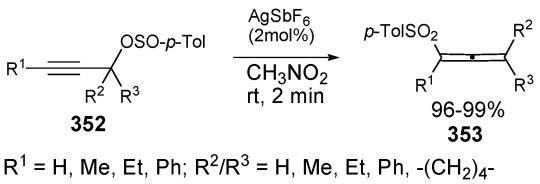
Scheme 117



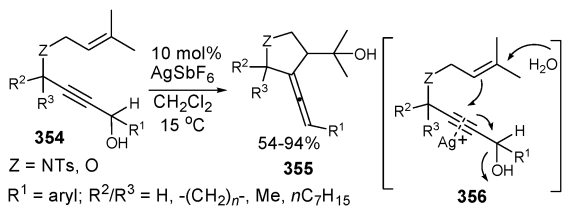
Scheme 118



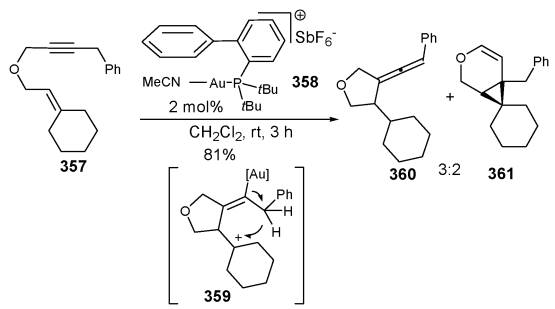
Scheme 119



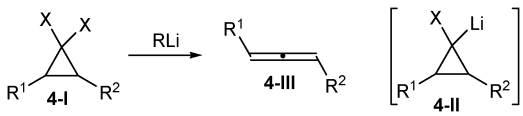
Scheme 120



Scheme 121



Scheme 122

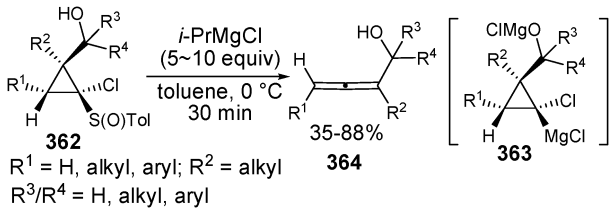


Scheme 123

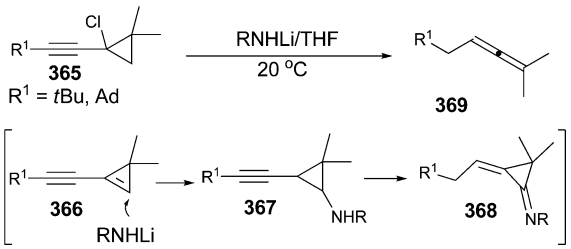
treated with a Grignard reagent, α-allenol **364** was obtained *via* the rearrangement of the cyclopropylmagnesium carbenoid intermediate.[227]

The reaction of lithium monoalkylamides with alkynylchlorocyclopropanes **365** bearing bulky substituents at the triple bond gave allenes **369** in 40–50% yields (Scheme 125).[228] This reaction was proposed to proceed *via* alkynylcyclopropene intermediates **366**.

Ethynylcyclopropane and its derivatives constitute versatile building blocks with enhanced reactivity across the triple bond and three-membered ring.[229] When propargylcyclopropane-1,1-diesters **370** were treated with organocuprate reagents in ether at room temperature, a regioselective ring opening of the cyclopropane took place to furnish allenes **371** with methyl malonate tethered at β-position (Scheme 126).[230] When chiral

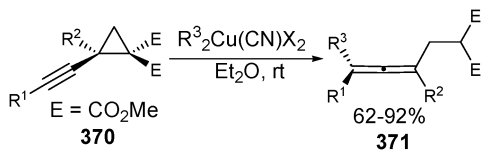


Scheme 124



Scheme 125

substrates were employed, enantioenriched allenes could be obtained. The use of ''higher order'' cuprates was critical in obtaining the reported results.



Scheme 126

Methylenecyclopropanes (MCPs) **372** underwent AgOTf-catalyzed ring expansion providing a diastereoselective access to polysubstituted allenylcyclobutanols **376** (Scheme 127).[231]

5. From allenes

Modification of suitable allenyl precursors provides a variety of flexible methods for some otherwise not easily available functionalized allenes.[232] For example, reduction of allenyl aldehydes or ketones with $LiAlH_4$, $NaBH_4$, DIBAL-H, and L-Selectride gives primary or secondary allenyl alcohols with high chemo- and regioselectivity.[54b,233] Metathesis reactions of monosubstituted allenes in the presence of the Grubbs carbene complex afford 1,3-disubstituted allenes in varying yields.[234] Allenylphosphine-boranes can be prepared by addition of borane to the corresponding allenyl phosphine.[235]

5.1 By transition-metal catalyzed cross-coupling reactions

Metalated or halogenated allenes **5-I** undergo transition-metal catalyzed cross-coupling reactions with electrophiles or organometallic species **5-II** respectively, leading to highly functionalized allenyl products **5-III** (Scheme 128).[1f,236] The easy access to starting material (see Section 2.2.1 for the preparation of haloallenes) makes this method especially attractive for C–C or carbon–heteroatom bond formations.

In 2005, Trost[237] and Hsung[238] reported the synthesis of *N*-substituted allenes by copper-catalyzed coupling reactions between allenyl halides and N-containing compounds, respectively. Recently, Bäckvall *et al.* examined the reaction of allyl- or aryl-substituted sulfonamides **378** with bromoallenes **377** in the presence of copper(I) thiophene-2-carboxylate (CuTC) (Scheme 129).[239] Allenic amides **379** were afforded in moderate to good yields.

Treatment of allenyl bromide **380** with propargylzinc, prepared from lithiation of 1-triethylsiloxyl-2-propyne and metal exchange with anhydrous $ZnBr_2$, gave allenyne **381** in 90% yield and a moderate ee (63%) (Scheme 130).[240] It was found that the absolute configuration of axial chirality was inverted during this transformation.

An allenylzinc reagent generated by lithiation of 1-aryl-3-alkylpropadienes **382** and subsequent transmetalation with zinc bromide coupled with sp^2 organohalides affording the corresponding trisubstituted allenes **383** in a highly regioselective fashion with moderate to excellent yields (Scheme 131).[241]

A stereocontrolled coupling of haloallenes **384** with boronic acids **385** was reported by Burke *et al.* (Scheme 132).[242] The success of this reaction was attributed to the use of sterically bulky phosphine ligands and allenyl iodides bearing steric bulk substituents at C3.

Palladium-catalyzed allene-type asymmetric allylation alkylation reactions are also known. Since 2002, Murahashi, Imada and Naoto *et al.* have disclosed a series of palladium complex catalyzed asymmetric allenylations.[243] Trost *et al.* developed a dynamic kinetic asymmetric allenylic alkylations of racemic allene acetates to give allenes with high enantiomeric excess (84–95%) for both malonate and amine nucleophiles.[244] Recently, Naota and Imada reported a one-pot asymmetric bishomoallenylation of primary amines **388** with 2,3-allenyl phosphates **387** catalyzed by a palladium/chiral phosphine complex. The reaction provided a facile and convenient method for synthesizing optically active bis(homoallenyl)amines **389** bearing two chiral axes (Scheme 133).[245]

Hamada *et al.* examined the reaction of 2,3-allenyl acetate **390** with α-substituted malonates **391** by using a chiral Pd-DIAPHOX catalyst system. The reaction proceeded in the presence of a catalytic amount of lithium acetate at 4 °C, affording the chiral allenes (*R*)-**393** in excellent yield with up to 99% ee (Scheme 134).[246]

5.3 From 2,3-allenonates to allenyl alcohols or ketones

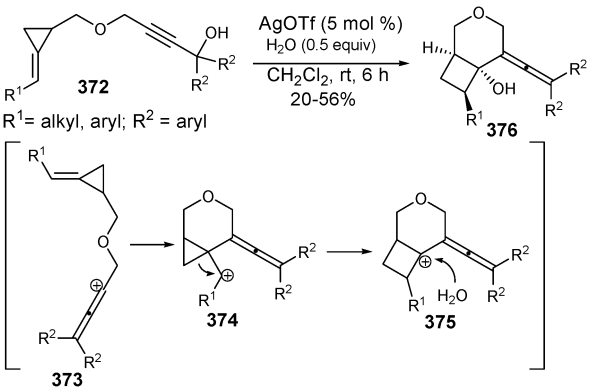
Previously, we have reported that 2,3-allenoates reacted with alkyl, aryl or alkenyl Grignard reagents in the presence of an iron catalyst to provide β,γ-unsaturated alkenoates with excellent stereoselectivity.[247a,b] Following this research, we found that 2,3-allenoates **394** could undergo a double 1,2-addition reaction with allyl magnesium chloride **395** in the absence of any transition metal catalyst to give tertiary α-allenols **396** (Scheme 135).[247c] Optically active allenol **398** could be prepared from the optically active 2,3-allenoate **397** without obvious racemization of the axial chirality.

We also found that when perfluoroalkyl Grignard reagents **399** were used, perfluoroalkyl allenyl ketones **401** could be obtained without further addition of R_fMgX to these perfluoroalkyl ketones (Scheme 136).[248]

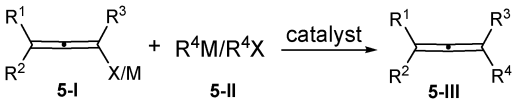
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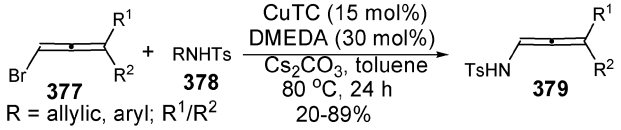
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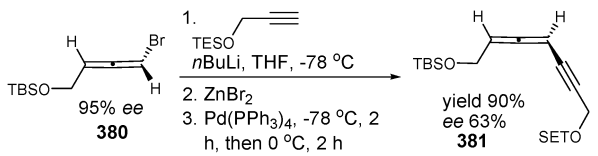
Scheme 127



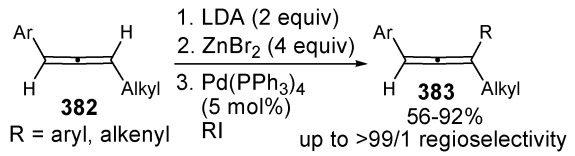
Scheme 128



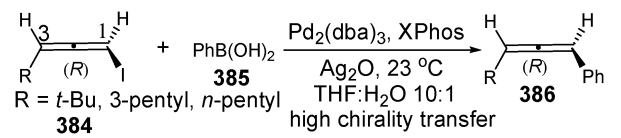
Scheme 129



Scheme 130



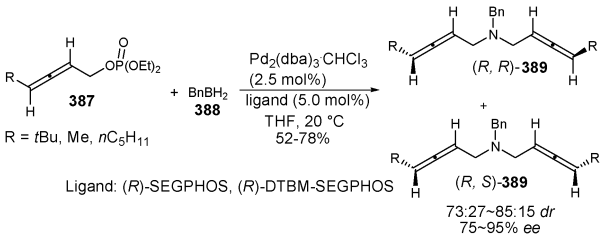
Scheme 131



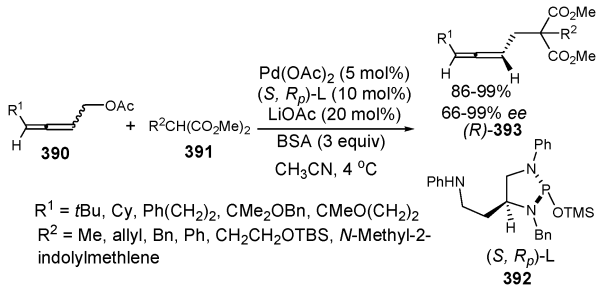
Scheme 132

Alkynyl enolates[140] formed from allenoates **402** under the mediation of TBAF added to aryl aldehydes **403** at the γ-position, yielded the corresponding α-carbinol allenoates **404** in moderate to excellent yields (Scheme 137).[249] With increasing steric repulsion at γ-position, an α-selectivity could also be achieved.

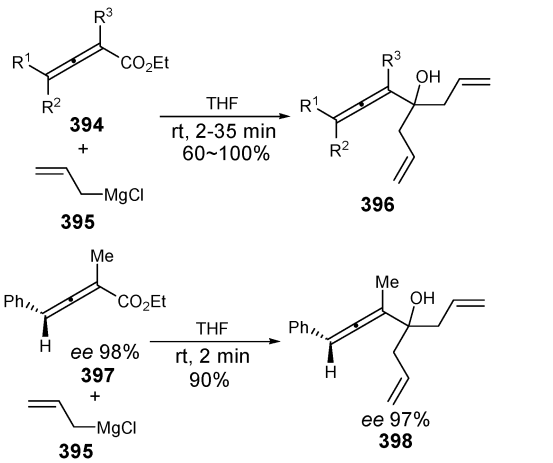
Organotrifluoroborate salts have emerged as versatile coupling partners showing superior reactivity and stability.[250]



Scheme 133



Scheme 134



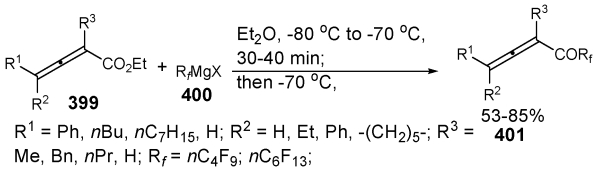
Scheme 135

Recently, Aggarwal *et al.* developed a method for the preparation of allenyl trifluoroborate **407** from the corresponding pinacolboronic ester by azeotropic removal of pinacol from the reaction mixture (Scheme 138).[251]

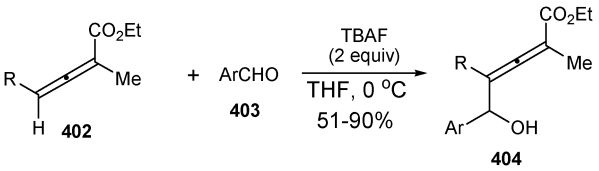
5.4 By resolution of allenic compounds

With increasing emphasis on enantioselective catalysis, optical resolutions have been perceived to be less elegant. However, resolution of racemic mixtures by use of inexpensive stoichiometric resolving agents or enzyme catalysis is still the method of choice when it comes to the bulky production of optically pure material where no reasonable enantioselective method is available.[252]

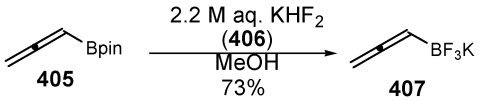
Optical active allenoic acids are an important class of functionalized allenes that serve as highly versatile synthetic intermediates.[1i,21c,61a,253] The cyclization of allenoic acids furnishes functionalized chiral lactones such as butenolides[254] and β-halobutenolides.[21d,143b] We have also found that



Scheme 136



Scheme 137



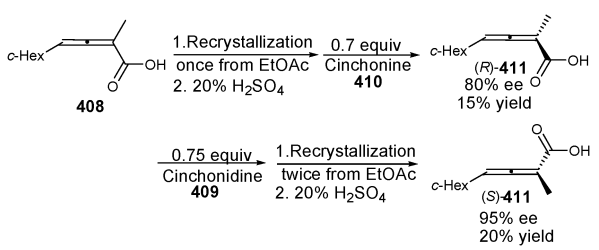
Scheme 138

acid–base salts of (+)-cinchonidine or (−)-/(+)-α-methylbenzylamine and 2,3-allenoic acids in an aqueous medium under the mediation of CuX_2 giving optically active β-halobutenolides.[255]

Recently, Bäckvall *et al.* reported an optical resolution of trisubstituted allenoic acid, 4-cyclohexyl-2-methyl-buta-2,3-dienoic acid **408**, by cinchonidine **409** and cinchonine **410**, respectively (Scheme 139).[256] After recrystallization, (+)-allenoic acid (*S*)-**411** was obtained in 95% ee and (−)-allenoic acid (*R*)-**411** was obtained in 80% ee.

Allenols are another important class of functionalized allenes,[116b,c,257] which have been used for the synthesis of oxiranes,[258] 2,5-dihydrofurans,[259] α-methylenelactones,[257c,260] α- or γ-amino alcohols,[116c] and α,β-unsaturated ketones.[261] Previously, we have reported that Novozym-435 (a form of *Candida antarctica* lipase B) is an efficient biocatalyst for the kinetic resolution of racemic 2,3-allenols, β-allenols and aryl-substituted propargylic alcohols.[233b,262] We have also found that PPL (porcine pancreatic lipase)-catalyzed kinetic resolution of 2-methyldeca-2,3-dienol **412** with vinyl acetate **413** gave alcohol **415** in 19% yield and >98% ee (Scheme 140).[263]

Meanwhile, Bäckvall *et al.* have found that in the porcine pancreatic lipase-mediated kinetic resolution of primary allenic alcohols **416**, the substitution patterns determined the enantioselectivities (Scheme 141).[264] Allenols with a substituent at the 2-position showed higher enantioselectivities together



Scheme 139

with a faster reaction rate, whereas alkenyl- and alkylsubstituted allenols displayed a considerably reduced enantioselectivity and to achieve highly enantioenriched material, an attentive over-acylation must be required.

Kinetic resolutions of racemates using asymmetric catalysts are less developed until recently. Gong reported a bisphosphoric acid catalyzed kinetic resolution of racemic 2,3-allenoates **419** *via* 1,3-dipolar cycloaddition (Scheme 142). Optically active 2,3-allenoates **423** with (*R*)-configuration and 3-methyl-enepyrrolidine derivatives **424** were afforded in high yields and enantioselectivities, respectively.[265]

The combination of transition-metal catalyzed racemization and enzyme catalyzed kinetic resolution to provide a powerful approach for obtaining enantiomerically pure compounds, where 100% yield becomes possible. Bäckvall *et al.* applied an NHC–palladium catalyst together with porcine pancreatic lipase to dynamic resolute allenols **425** giving allenyl butyrates (*R*)-**427** in good to high yields and in *ee* values up to 89% (Scheme 143).[266]

6. Total synthesis of allenic natural products

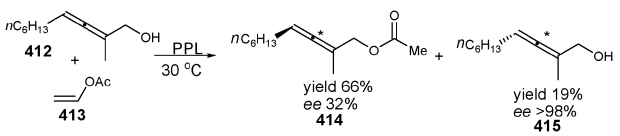
There are more than 150 allenic natural products reported to date.[3a,61a,267] Many of them are chiral and show interesting biological activities. In this part, only the recent examples on chemical synthesis of allenic natural products will be discussed with the emphasis on the generation of allene moiety.

6.1 Phomallenic acid C

Phomallenic acid C **432** was isolated from the fermentation broth of *Phoma sp.* as potent inhibitors of the type II fatty acid synthesis (FAS II) pathway.[268] It featured a linear allenyldiyne structure and the absolute configuration of allene moiety was supposed to be *R* according to Lowe's rule. Due to its unique structure as well as significant biological activity, several groups have conducted the total synthesis of this compound in either racemic or optically active forms. All of them employed transition-metal catalyzed cross-coupling reactions to construct the allenyldiyne moiety. In 2007, Wu *et al.* reported the first stereoselective synthesis of phomallenic acid C **432** (88% ee). He applied the Negishi coupling of diyne **430** and allenyl bromide **429** as a key step in this enantioselective synthesis (Scheme 144).[269]

Recently, Yoshida *et al.* reported a synthesis of racemic phomallenic acids C **432** by using palladium-catalyzed coupling of propargylic tosylates **433** with terminal alkynes **430**. Attempts to the asymmetric synthesis utilizing optically active propargylic compounds failed due to uncontrollable racemization (0% ee) (Scheme 145).[270]

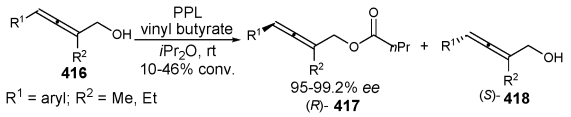
A highly enantioselective synthesis of phomallenic acid C (*R*)-**432** was disclosed by Ishigami *et al.* recently (Scheme 146).[271] The key allenyldiyne structure was



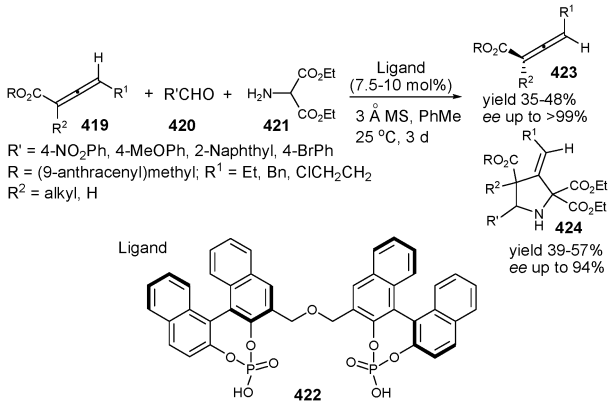
Scheme 140

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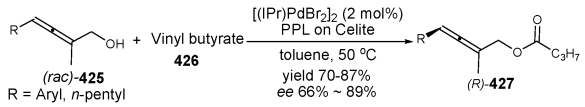
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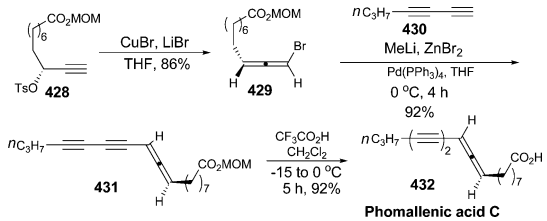
Scheme 141



Scheme 142



Scheme 143

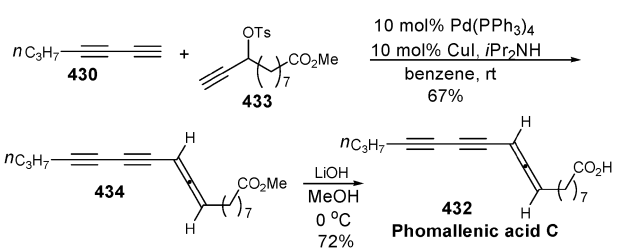


Scheme 144

constructed in a 98% ee by a direct anti-S_N2' coupling of propargyl mesylate (*S*)-**435** with indium(II) diacetylide in the presence of palladium catalyst. The total yield was 23% in eight steps from the known aldehyde, 9-*tert*-butyldimethylsilyloxy nonanal.[272]

6.2 (−)-Marasin

(a*R*)-(−)-Marasin (*aR*)-**443**, isolated from *Marasmius ramealis* and *Cortinellus berkeleyanus*, and (a*S*)-(+)-Marasin (*aS*)-**444** from *Aleurodiscus roseus* feature an allenediyne structure showing bioactivity against numerous bacteria including *Mycobacterium tuberculosis*. Graaf *et al.* reported the first synthesis by using a ferrocene derived chiral catalyst-mediated coupling of a zincated allene with a bromodiyne as the key step and eventually resulted in (*aR*)-(−)-marasin in 0.5% ee.[273] A biosynthetic way was developed by Davies and Hodge in 2005.[274]



Scheme 145

Recently, Wu *et al.* explored the synthetic studies toward the allenediyne antibiotic marasin and achieved an much higher enantioselectivity than previous reported chemical synthesis (Scheme 147).[275] Several different approaches to the construction of the optically active chiral allenediyne motif were examined. Among these, coupling of an optically active bromoallene (*aS*)-**438** (78% ee, prepared from the corresponding propargyl tosylate **437**) with a zincated diyne **439** gave the (*aR*)-(−)-allenediyne (*aR*)-**440** in 36% *ee*. Direct coupling of the tosylate precursor **437** (97% ee) with the same diyne species led to the allenediyne (*aS*)-**440** of higher enantiopurity (89% ee), but of different axial configuration.

Installation of the same allendiyne moiety through elimination of a diyne-containing allyl acetate–vinyl sulfoxide **441** (98% ee) afforded the allendiyne (*aS*)-**442** in 73% ee, with a chirality transfer efficiency of 74% (Scheme 148).

When the chiral sulfinyl was replaced with iodine in the elimination step, the chirality was more efficiently transferred from the allylic acetate **445** to the allenic axis, with the efficiency being 84% (Scheme 149).

If the substrate for the elimination reaction contained one less triple bond, the corresponding allenyne (*aS*)-**447** could be built with much better enantiopurity (94% *ee*) (Scheme 150). However, subsequent attachment with another acetylene unit *via* alkyne–alkyne coupling failed due to the easy decomposition of the substrate.

6.3 Scorodonin

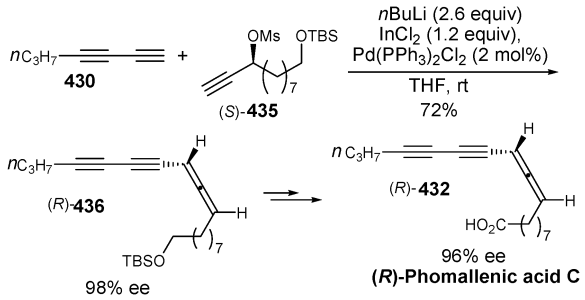
Scorodonin was isolated from *Marasmius scorodonius* with inhibiting effects on the growth of bacteria, yeasts and filamentous fungi.[276]

Recently, Wu *et al.* reported the first synthesis of this natural product (*aR*)-**453**, with the key allenic axis formed in 63% ee *via* the coupling reaction between an optically-active bromoallene (*aR*)-**449** (95% ee) and a zincated terminal alkyne **448** (Scheme 151).[240] The enantiopurity of natural scorodonin was calculated as about 89% or lower from the optical rotation of synthetic sample. Such a phenomenon might be common to all similar natural allenic products.

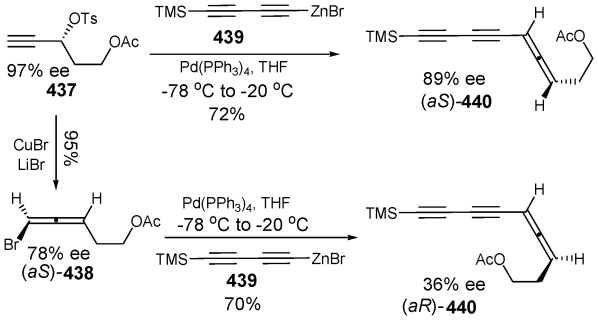
Due to the inefficiency of chirality transfer, an elimination approach was applied to construct the allenyne moiety (Scheme 152).[277] Starting from the (*S*)-sulfoxide (*S*)-**454** of 88.5% *ee*, (a*S*)-allenyne (*aS*)-**456** was obtained in 87% *ee*. The stereoselectivity was substantially higher than that (63% *ee*) generated by bromoallene-alkyne coupling approach. It has been well established that the chirality of the newly formed allenic axis is dictated by the allylic stereogenic center. However, Wu *et al.* found that the relative configuration of the sulfinyl group appeared also showing a substantial influence.

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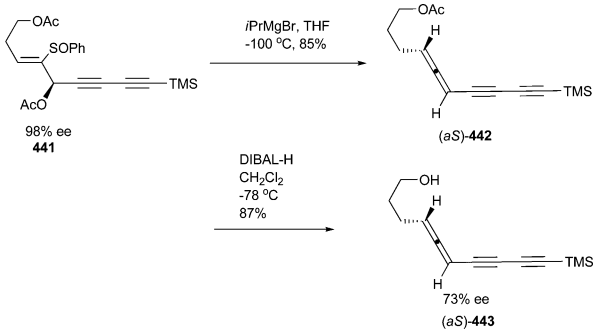
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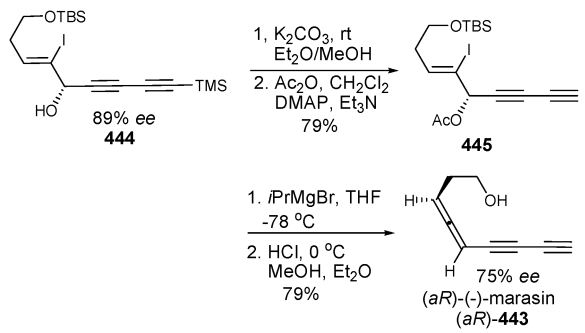
Scheme 146



Scheme 147

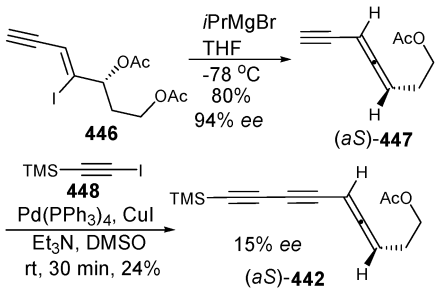


Scheme 148

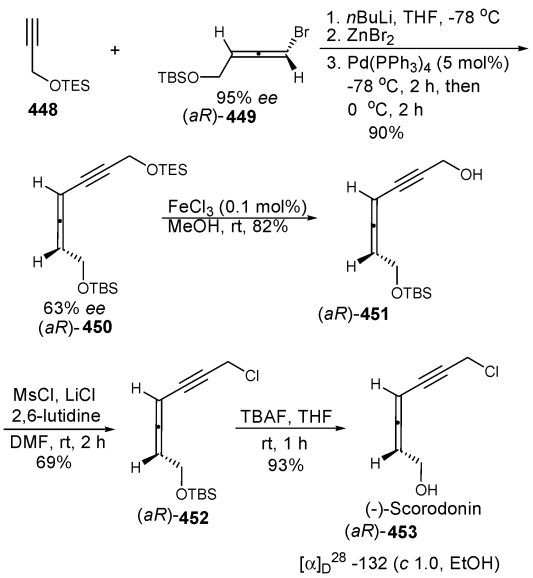


Scheme 149

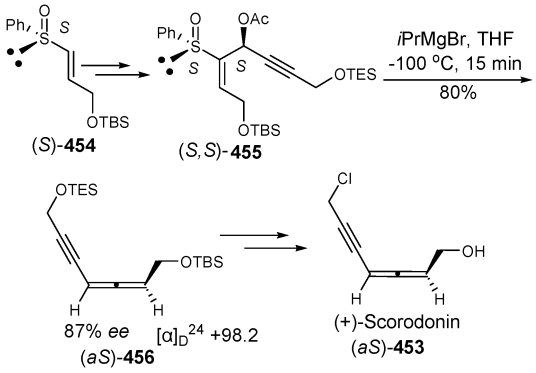
When the chirality of the sulfinyl group "matched", the chirality at the allylic stereogenic center could be efficiently transferred to the allenic axis. The efficiency of the chirality transfer for sulfinyl group (*S*) with a (*R*) configuration at the allylic position, was much lower (48% *ee*) than that with



Scheme 150



Scheme 151



Scheme 152

sulfinyl group (*S*) with a (*S*) configuration at the allylic position (87% *ee*).

6.4 Nemotin

Nemotin was isolated from fungi by Robbins and co-workers in the 1940s[278] and later was determined to have a "planar" structure based on UV, IR, hydrogenation and elemental analysis data.[279] The preliminary biotesting showed that it possessed significant activity against a range of microorganisms.

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The allene–diyne natural product nemotin **459** was synthesized through an enantioselective route with the stereogenic center at the lactone moiety derived from L-glutamic acid (Scheme 153).[280] The allene axis was constructed from the corresponding propargylic tosylate **457** by the palladium catalyzed Negishi coupling reaction. From this study, the relative as well as absolute configuration of this natural product was thus established to be (4*S*, 5*aS*).

6.5 (±)-Panacene, (±)-epipanacene, (±)-desbromopanacene and (−)-aplysiallene

(±)-Desbromopanacene was a plant metabolite isolated in 1915 from *Panax ginseng* and *Panax quinquefolius*.[281] Panacene was isolated in 1977 by Meinwald and co-workers from *Aplysia brasiliana*, a sea hare indigenous to the gulf coast of Florida with shark antifeedant properties.[282] These compounds feature a tri-fused cyclic benzofuran core with an allene pendant.

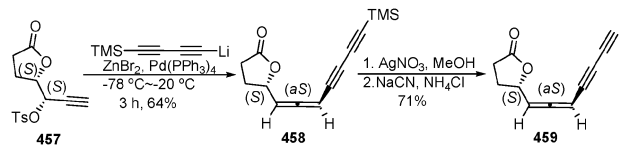
Feldman and Boukouvalas reported the total synthesis of racemate[283] and the natural (−)-form of Panacene (−)-**461**[284] in 1982 and 2006, respectively. Both of them applied the copper-mediated anti-S_N2' substitution of propargyl trisylates or mesylate **460** with bromide to construct the bromoallene moiety (Scheme 154).

Recently, Canesi *et al.* reported a concise diastereoselective synthesis of the marine (±)-panacene **461** and terrestrial (±)-desbromopanacene **458** based on an iodobenzene diacetate-mediated oxidative [2 + 3] cycloaddition between a substituted phenol and furan to build the *tri*-fused cyclic benzofuran core. Three strategies were employed to synthesis of the allene segment.[285] Intramolecular oxymercuration of bromo-substituted conjugated enyne **462**, followed by thiol-mediated configuration retentive proton demercuration of α-mercurio carbonyl compounds produced the correct configuration of the bromoallene moiety for (±)-panacene **461** (Scheme 155).

Bromoetherification of enynes **464** furnished the bromoallene moiety of (±)-epipanacene **465** highly regioselectively by using 2,4,4,6-tetrabromo-2,5-cyclohexadienone (TBCD) **466** in apolar solvents (Scheme 156). TBCD was more selective than *N*-bromosuccinimide as the latter promoted aromatic ring bromination.

The unsubstituted allene present in (±)-desbromopanacene **468** was efficiently introduced through Lewis acid-mediated nucleophilic addition of five-membered-ring oxocarbenium ions with propargyltrimethylsilane (Scheme 157).

Aplysiallene was isolated from the red alga *Laurencia okamurai Yamada*[286] and the sea hare *Aplysia kurodai*.[287] It features a unique *cis*-fused 2,6-dioxabicyclo[3,3,0]octane skeleton, an unusual (*E*,*E*)-bromodiene side chain, and an



Scheme 153

R-bromoallene appendage. Pagenkopf *et al.* reported the first total synthesis of aplysiallene starting from (*S*,*S*)-diepoxybutane in 16 steps with 2.3% overall yield and reassigned the structure of natural product to be (−)-aplysiallene **470** by comparison of the NMR data and optical rotation with the original one.[288] In their synthetic route, the *R*-bromoallene substituent was prepared by treating the corresponding *R*-propargyl trisylates **469** with $LiCuBr_2$.(Scheme 158).

6.6 (−)-Histrionicotoxin 285A

(−)-HTX 285A, the most abundant and potent[289] alkaloid within the histrionicotoxins (HTXs) family,[290] containing a terminal *cis*-enyne and allene side chain.

Holmes *et al.* reported the first total synthesis of (−)-histrionicotoxin 285A (−)-**475** starting from commercially available (*S*)-glycidol **471** (Scheme 159).[291] The six-membered chiral cyclic core was constructed by nitrone dipolar cycloaddition. The terminal allene moiety was formed by S_N2 substitution of alkyl iodide **472** with an excess of allenyl lithium in the presence of HMPA. The reaction was remarkably clean and concomitantly cleaved the triisopropylsilyl protecting group to afford the allene derivative in excellent yield.

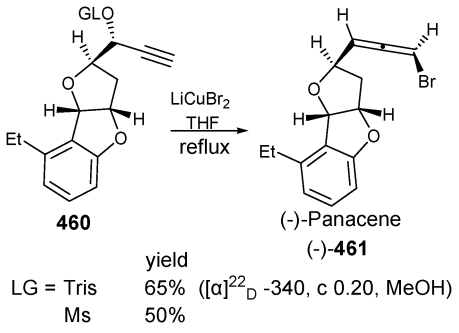
6.7 Peridinin and paracentrone

Peridinin, the most abundant photosynthetic pigment, was from planktonic dinoflagellates. Paracentrone was isolated from the sea urchin *Paracentrous lividus*. Their highly conjugated polyenes structure was found to play fundamental roles as photoprotective and antioxidation agents.[292]

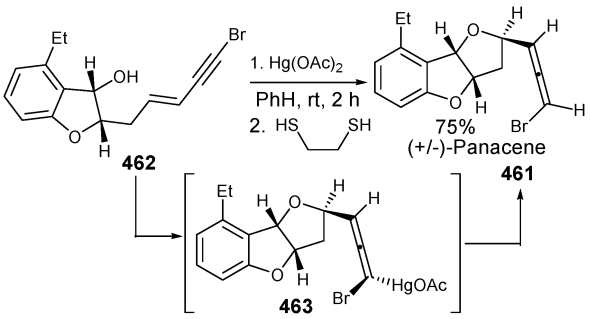
In 1990, Ito *et al.* described the first total synthesis of enantiopure peridinin **482**, who applied an S_N2'-reduction of the propargylic oxirane with diisobutylaluminum hydride to synthesize the allene moiety.[293] The *syn*-diastereoselectivity was ascribed to the precoordination of the aluminum hydride to the epoxide oxygen atom. Subsequently, this methodology was employed by Katsumura and co-workers in their stereo-controlled total synthesis of peridinin[294] and its derivatives.[295] Recently, de Lera *et al.* investigated the synthesis of peridinin with the expect to use the Stille cross-coupling to construct the allenene moiety *via* the coupling of bromoallene **477** with vinylstannane reagent **478** (Scheme 160).[296] It was found the configuration at the allene centre was inversed during the transformation. Thus, an epimer of peridinin was prepared followed this route. In the end, they turned to the well established S_N2'-reduction of the propargylic oxirane **480** with diisobutylaluminum hydride to get the allene part with right stereochemistry.

The first synthesis of paracentrone was achieved by Haugan *et al.* in 1996 over 13 linear steps from the readily available (3*R*,5*R*)-actinol.[297] Katsumura *et al.* used the similar strategy, as mentioned above, preparing the allenene fragment from acetylene **483** and trienyl vinyl iodide **484** by employing the Sonogashira coupling and S_N2'-hydride reduction (Scheme 161).[298] This allenene fragment was then taken on to the synthesis of paracentrone **487**.

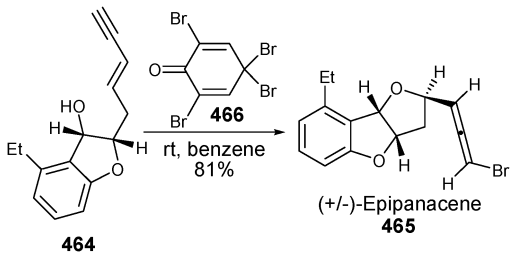
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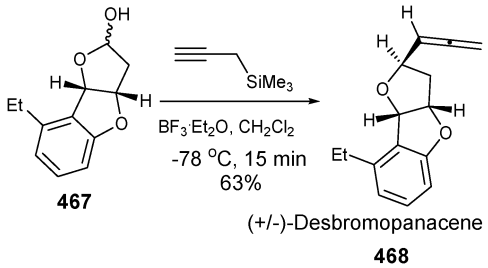
Scheme 154



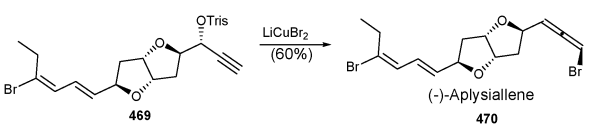
Scheme 155



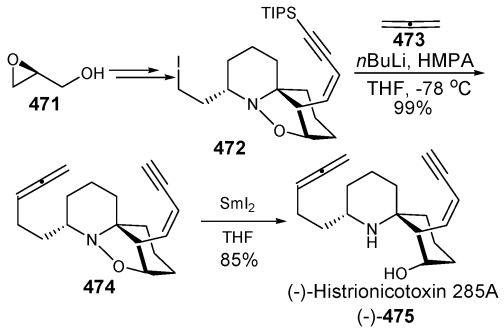
Scheme 156



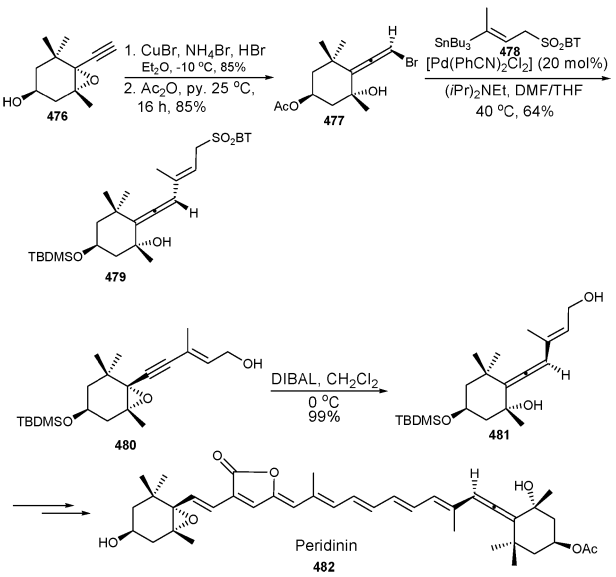
Scheme 157



Scheme 158



Scheme 159



Scheme 160

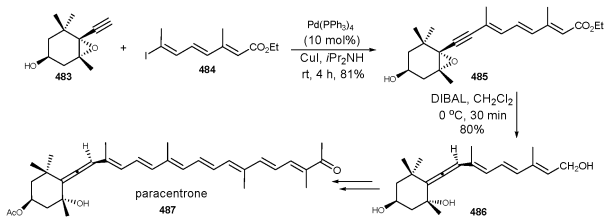
7. Miscellaneous

7.1 Synthesis of allene-based ligands

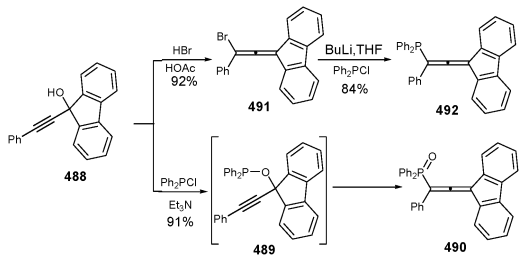
Although allenic phosphanes, phosphites and phosphates as valuable reaction substrates have been explored, they are less used as ligands partly due to the general consideration that they are maybe not stable during the reaction process. Still, several ligand systems based on allenes have been reported.

Treatment of 9-(phenylethynyl)fluoren-9-ol **488** with HBr in HOAc generated bromoallenes **491**, which reacted with butyllithium and chlorodiphenylphosphine furnishes 3,3-(biphenyl-2,20-diyl)-1-diphenylphosphino-1-phenylallene **492** (Scheme 162).[299] The reaction of 9-(phenylethynyl)fluoren-9-ol **488** with chlorodiphenylphosphine in the presence of Et_3N gave the corresponding allenylphosphine oxide **490** *via* [2,3]-sigmatropic rearrangements in high yield. Organometallic chemistry studies revealed that allenylphosphine reacted with transition metal such as $Fe_2(CO)_9$, $(THF)Cr(CO)_5$, $[(C_6H_5R)RuCl_2]_2$, initially as a conventional mono-phosphine ligand. But, when a readily displaceable ligand was available at the metal centre, the allene double bond adjacent to the phosphorus could also function as a donor.

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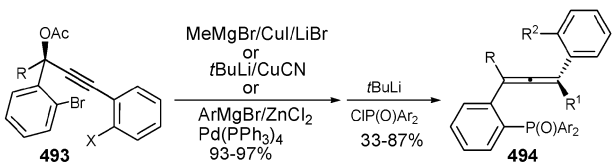


Scheme 161



Scheme 162

The first transition metal/allenic phosphine ligands mediated asymmetric transformation was reported by Ready *et al*. (Scheme 163).[300] The allenic phosphine oxides **494** were prepared by organocuprate-mediated or palladium-catalyzed propargylic substitution, followed by bromo-lithium exchange and trapping the resulting lithium intermediate with diarylchlorophosphine oxides. These ligands were subjected to the $SnCl_4$-mediated ring opening of *meso*-epoxides and showed a high chiral induction (82–94% ee).



R = Me, C_6H_5; R^1 = Me, *t*Bu, C_6H_5; R^2 = H, $P(O)Ar_2$;
Ar = C_6H_4, *p*-Tol, *p*-$CF_3C_6H_4$, 3,5-$(CH_3)_2C_6H_3$, 3,5-$(CF_3)_2C_6H_3$
X = H, Br

Scheme 163

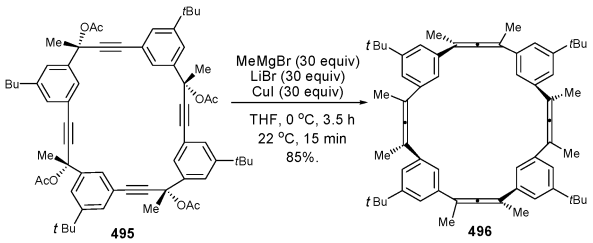
7.2 Synthesis of allenophane macrocycles

Allenophane macrocycles with large number of π-conjugated C atoms and exceptional chiroptical properties showed outstanding promise as optoelectronic materials and as strength-increasing additives to composite materials.[301] In addition, these molecules are useful as chiral ligands or as hosts for metal ions and small guest molecules.[302] Since Krause reported the first synthesis of a allenophane in 1999,[303] a number of allenophane macrocycles have been reported.[302]

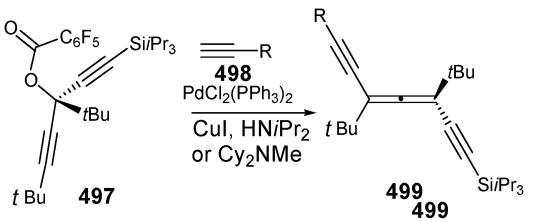
Recently, Fallis *et al*. reported the synthesis of a series of enantiomerically pure allenophanes starting from chiral epoxide.[304] The allene skeleton was prepared by organocuprate-mediated S_N2' coupling reaction of optical active propargyl acetates **495** (Scheme 164).

Optically active 1,3-diethynylallenes (DEAs)[305] were developed by Diederich *et al*. to synthesize chiral allenophanes without aromatic substituents.[306] This intermediate **499** was prepared by enantioselective Pd-mediated S_N2'-type cross-coupling of bispropargylic pentafluorobenzoate **497** and alkynes **498** (Scheme 165). In the course of this study, they found that the enantioselectivity was very sensitive to the nature of the base. Two enantiomers were obtained selectively by using $CyNMe_2$ or iPr_2NH as base. X-ray analysis revealed that Pd-mediated S_N2'-type substitution proceeded *via* a *syn* manner, which was in sharp contrast to the normal observed *anti* substitution.

With this newly established methodology, novel chromophores with interesting optical properties, such as **504**, were now available *via* simple transition metal-catalyzed cross-coupling (Scheme 166).

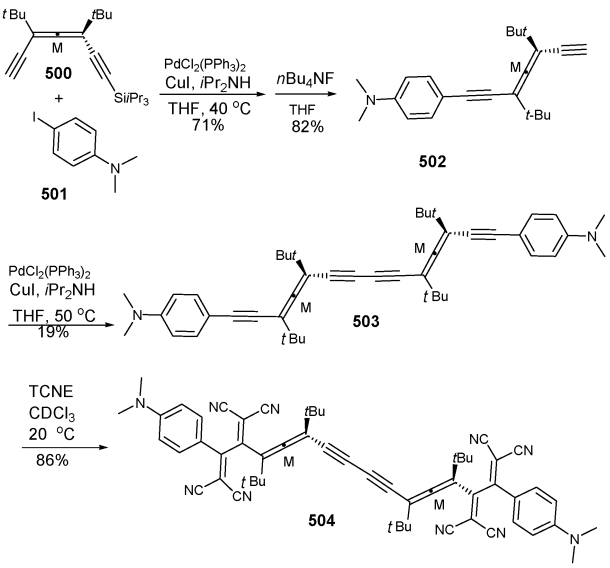


Scheme 164



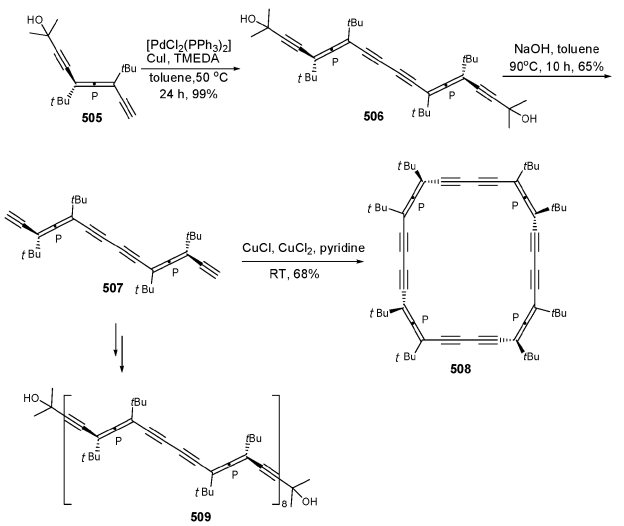
R	base	T [°C]/t [h]	Yield [%]/e.r. M:P
TMS (*S*)	Cy_2NMe	30/5	61/90:1
TMS (*R*)	iPr_2NH	20/20	90/5:95
CMe_2OH (*S*)	Cy_2NMe	30/96	69/96:4
CMe_2OH (*R*)	iPr_2NH	20/96	49/7:93

Scheme 165



Scheme 166

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Scheme 167

The enantiomerically pure (*P*,*P*,*P*,*P*)-(-)-alleno-acetylenic macrocycle **508** was prepared following the three-step protocol (Scheme 167).[307] The enantiomerically pure (*P*)-(+)-DEA derivative **505** was dimerized by palladium-catalyzed oxidative homocoupling to yield linear (*P*,P)-(+)-alleno-acetylenic dimer **506** quantitatively. After deprotection, enantiomerically pure (*P*,*P*,*P*,*P*)-(−)-macrocycle **508** was obtained in a one-pot dimerization–cyclization reaction of compound (*P*,*P*)-(+)-alleno-acetylenic dimer **507**. A racemate alleno-acetylenic macrocycle bearing four 2,5-substituted pyridine units has been prepared by using the similar strategy. Four possible diastereoisomers (twist, chair, crown, and boat) were formed and they could be separated by preparative HPLC.[308] By repeating the dimerization–deprotection sequence without the cyclization step, linear enantiopure alleno-acetylenic oligomers **509** would be furnished.[309]

Conclusions

With these recent highlighted examples we hope to raise the reader's attention to this exciting chemistry and convinced him/her that allenes are no longer chemist curiosities. The applications of allenes in the total synthesis, medicinal chemistry and advanced materials will spur much interest into both the synthesis of allenes as well as their further applications. Great advances have been achieved with the development of new catalytic systems and more ligands available now. A variety of new methods for complex, multi-functionalized, especially chiral allenes have been developed. Revitalizing "the old" reactions leads to wider substrate scope, milder reaction conditions, finely-tuned regio- and stereoselectivities and even new synthesis for allenes. New sources of starting material have also been explored; nevertheless alkynes still dominate this territory. One problem not yet been fully resolved is the lack of proper and simple methods for supplying enantiomerically enriched compounds from readily available starting material. Currently, the synthesis of enantiomerically enriched allenes has relied on the transfer of chirality from a stereogenic center to the allene axis, optical resolution of racemic allenes, and recently, catalytic enantioselective synthesis of chiral allenes.[24,25] New approaches from readily available chemicals are highly desirable, which should lead to further applications of allenes in stereoselective syntheses, especially in those of natural products, pharmaceuticals, and material science. In the future, this point will probably constitute the most exciting challenge waiting for chemists to conquer.

Acknowledgements

We thank Prof. Chunling Fu in this group for checking all the references. We thank the State Basic Research and Development Program (Grant No. 2011CB808700) and National Natural Science Foundation of China (20732005) for financial support.

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